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Key Aroma Compounds and Aroma Characteristics in Nongxiangxing Baijiu Characterized by GC-IMS, GC × GC-MS, and an Electronic Nose

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ABSTRACT: Nongxiangxing (NXX) baijiu belongs to one of the major flavor categories of Chinese baijiu; however, the duration of storage is of vital importance in developing the distinctive flavor of baijiu, which may affect its aroma, taste, and body structure. In this research, the volatile compounds in five NXX baijiu (Y1, Y3, Y5, Y8, and Y11) from different storage years were qualitatively and quantitatively identified by gas chromatography-ion migration spectrometry (GC-IMS) and two-dimensional gas chromatography-combined mass spectrometry (GC × GC-MS). GC-IMS, orthogonal partial least-squares-discriminant analysis (OPLS-DA), and variable importance in projection (VIP) analysis identified 11 key aroma compounds; subsequently, relative odor activity value (ROAV ≥ 1) analysis was used to further determine 2 potential key aroma compounds. GC × GC-MS, OPLS-DA, and VIP analysis identified 16 key aroma compounds, and odor activity value (OAV ≥ 1) analysis was used to further determine 8 potential key aroma compounds. By integrating the electronic nose (E-nose) and key aroma compound analysis, the significant differences in aroma between samples were revealed, the aroma distribution between Y3 and Y5 was more similar, the aroma distribution between Y8 and Y11 was more similar, and the roasted aroma, nut aroma, fruit aroma, and ester aroma were more prominent in Y1. Moreover, the correlation analysis conducted between the response values of the E-nose sensors and the levels of some key aroma compounds detected by GC × GC-MS showed that the E-nose exhibited a strong discrimination ability for Y1.

KEYWORDS: *Nongxiangxing baijiu, key aroma compounds, aroma characteristics*

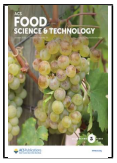
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INTRODUCTION

Based on different flavor characteristics, baijiu can be classified into 12 categories. These include Sauce-flavor baijiu, Strong-flavor baijiu (NXX baijiu), Light-flavor baijiu, Miscellaneous-flavor baijiu, Feng-flavor baijiu, Rice-flavor baijiu, Medicine-flavor baijiu, Sesame-flavor baijiu, Te-flavor baijiu, Chi-flavor baijiu, Laobaigan-flavor baijiu, and Fuyu-flavor baijiu. Among them, NXX baijiu accounts for over 70% of the baijiu industry and is the main baijiu consumer product in China. NXX baijiu has the characteristics of strong cellar aroma, harmonious aroma, soft and sweet, long tail clearance, and so on. The differences in brand, raw materials, pit age, production technology, and storage time of baijiu result in various flavors of the produced baijiu and make the volatile compounds in baijiu exhibit the characteristics of rich types and low content. These volatile compounds are crucial in determining the distinctive flavor, quality, and style of baijiu.

Unstored baijiu contains odors and irritants, such as hydrogen sulfide, mercaptan, thioether, free ammonia, and olefinic aldehyde, which may show the characteristics of dryness and spiciness in taste. It is necessary to achieve a new balance in a series of physical and chemical changes through storage so that its taste becomes soft and harmonious. In the production of mainstream baijiu products both at home and abroad, storage is an essential step. It can have a remarkable impact on the aroma quality and value of baijiu.[1] During the storage process of baijiu, the volatile chemicals with a low boiling point and unpleasant odors in baijiu are removed through evaporation and chemical reactions, including oxidation, esterification, and hydrolysis. This process facilitates the formation of the desired aroma compounds.[2] More precisely, the concentration of compounds including acids, esters, and alcohols varies during the storage period, which leads to an aroma style that is well-balanced.[3] Storage can also enhance freshly produced baijiu by removing pungent smells and improve the aging quality, making the baijiu taste soft and mellow.

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Gas chromatography-ion migration spectrometry (GC-IMS) serves as a method applied to quickly detect volatile components within samples. It integrates GC and IMS.[4] This technique works on the principle that after the complex compound components are separated by a column, the ionized compounds require different drift times (Dt) to travel a fixed distance (drift tube) in a particular electric field.[5] It has several advantages, such as high gas chromatographic separation efficiency, high sensitivity, and low cost. However, IMS has limitations in terms of low resolution when dealing with mixtures and compounds with similar mobility. Gas chromatography−mass spectrometry (GC-MS) is a powerful tool for the separation and identification of complex volatile compounds in the qualitative and quantitative analyses of volatile components in food. This technique offers the advantages of high resolution and sensitivity. However, it also has certain drawbacks. For example, the pretreatment process is intricate, and it is challenging to distinguish peak fragmentation. An E-nose is an important tool for electronic bionic evaluation. It simulates the human olfactory function through the interaction of different types of gas sensors with sample odor molecules, detects, analyzes, and recognizes complex odors and volatile components, and then realizes the analysis work of sample identification, classification, and quality control. It has the advantages of simple preprocessing, short detection cycle, and is suitable for large sample testing. Nevertheless, it has a low accuracy rate. The sensor tends to drift easily, and it is impossible to conduct a qualitative analysis of a single volatile compound. Taylor[5] found that samples can be differentiated by both GC-MS and GC-IMS. The employment of a combination of instruments has gained popularity. It is useful for isolating and identifying a broader spectrum of volatile compounds. Moreover, it can provide a more complete and diverse set of food flavor characteristics. These instruments complement one another in validating analytical outcomes and distinguishing samples in a more comprehensive and effective way. HS-GC-IMS and DI-GC-MS were utilized to explore the evolution rules of Xiangjiao Baijiu during storage. It was found that ethyl octanoate and another 10 compounds were crucial compounds causing the differences among Xiangjiao baijiu stored in different containers and from different years.[6] According to the research report,[7] the integration of HS-SPME/GC-MS, HS-GC-IMS, and E-nose could present sample flavor information more comprehensively and reliably. The overall flavor characteristics of baijiu were examined using E-nose, HS-SPME-GC-MS/MS, and HS-GC-IMS. It was discovered that the key volatile compounds in NXX baijiu were mainly featured by floral, fruity, grassy, and nutty aroma.[8] Although many studies have explored the impact of storage time on various types of baijiu, research that specifically focuses on the influence of storage time on the aroma characteristics of NXX baijiu is still scarce. Moreover, no studies have reported on the difference in the potential key aroma compounds in NXX baijiu during different storage periods.

To comprehensively explore the aroma characteristics of NXX baijiu, this research innovatively integrated GC-IMS, GC × GC-MS, and E-nose technologies. A thorough analysis of volatile compounds in Y1, Y3, Y5, Y8, and Y11 was conducted and the impact of different storage times (1, 3, 5, 8, and 11 years) on the aroma of NXX baijiu was investigated. GC-IMS was employed to explore the specific influence of the storage time on the fingerprint of volatile compounds in NXX baijiu. Potential key aroma compounds were recognized through the OPLS-DA model in combination with ROAV. GC × GC-MS was employed to analyze the volatile compounds existing in these NXX baijiu samples. Potential key aroma compounds were filtered out by integrating the OPLS-DA model and the OAV. Moreover, an E-nose was used to assess the effect of storage time on volatile components. Through the above comprehensive research, it was found that NXX baijiu was rich in a variety of volatile compounds, among which certain key aroma compounds were of great significance in the overall aroma and could distinguish the aroma difference among the five types of baijiu. The storage time is of vital and decisive significance in shaping the composition of volatile compounds and aroma characteristics. These research results offer a theoretical foundation and technical guidance for improving the quality and aroma of NXX baijiu by controlling the storage time.

MATERIALS AND METHODS

Materials and Samples. 2-Octanol (99.5%) and *n*-ketones (C4−C9) purity > 99% from Sigma-Aldrich (Beijing, China).

Five different storage-year NXX baijius were analyzed, including Y1 (1 year, 52% vol), Y3 (3 years, 52% vol), Y5 (5 years, 52% vol), Y8 (8 years, 52% vol), and Y11 (11 years, 52% vol). All samples were from exactly the same batch of baijiu and were industrially manufactured by a renowned distillery located in Chengdu City, Sichuan Province, China. The distillery used the same brewing materials, distillation techniques, fermentation conditions, and storage environments during production. The samples were stored in a well-packaged state at 25 °C until they were ready for further analysis.

GC−IMS Analysis of Volatile Compounds. GC-IMS described by Cai et al.[9] was used to analyze each sample, and slightly modified according to the actual situation. In short, 0.2 mL of the NXX baijiu sample was taken, diluted with 1.8 mL of water, placed in a 10-mL headspace bottle, and the bottle was kept at 60 °C (500 rpm) for 10 min. 500 μL of the sample was taken from the headspace bottle and injected into GC-IMS, the temperature of the syringe was set at 65 °C during the injection process. A capillary column, MXT-WAX (30 m × 0.53 mm, 1 μm) (Restek, United States) was employed. The temperature of the gas chromatography column was adjusted to 60 °C. The carrier gas employed was high-purity nitrogen (purity ≥ 99.999%). The flow rate of the carrier gas was configured in the following manner: Initially, it was set at 2 mL/min and sustained for 2 min. Subsequently, it increased linearly at a gradual pace until it reached 10 mL/min. Finally, it was linearly increased to 100 mL/min and maintained at this level for 10 min. The overall analysis process lasted for 20 min. Using a 3H ionization source in an IMS ionization chamber, volatile compounds underwent ionization in the positive ion mode. Subsequently, the volatile compounds were introduced into a 98 mm drift tube. The drift gas employed was high-purity N_2, and it had a flow rate of 150 mL/min at a temperature of 45 °C. The retention index (RI) was computed by employing the positive ketone series (C4−C9). To identify the volatile compounds, they were compared with the qualitative software GC-IMS library retrieval (which contained the pre-installed NIST 2014 database and IMS database). The relative percentage content of each volatile component was computed through the peak intensity normalization method. The contribution made by the volatile component to the overall flavor was analyzed through ROAV. The calculation formula of ROAV was presented as follows

$$\mathrm{ROAV}_x = 100 \times \left(\frac{C\%_x}{C\%_{\mathrm{Stan}}}\right) \times \left(\frac{T_{\mathrm{Stan}}}{T_x}\right)$$

ROAV_x represents the relative odor activity value of a volatile component, while $C\%_x$ and T_x respectively denote the relative percentage content and threshold of volatile component. $C\%_{\mathrm{Stan}}$ and T_{Stan} represent the relative percentage content and threshold of the

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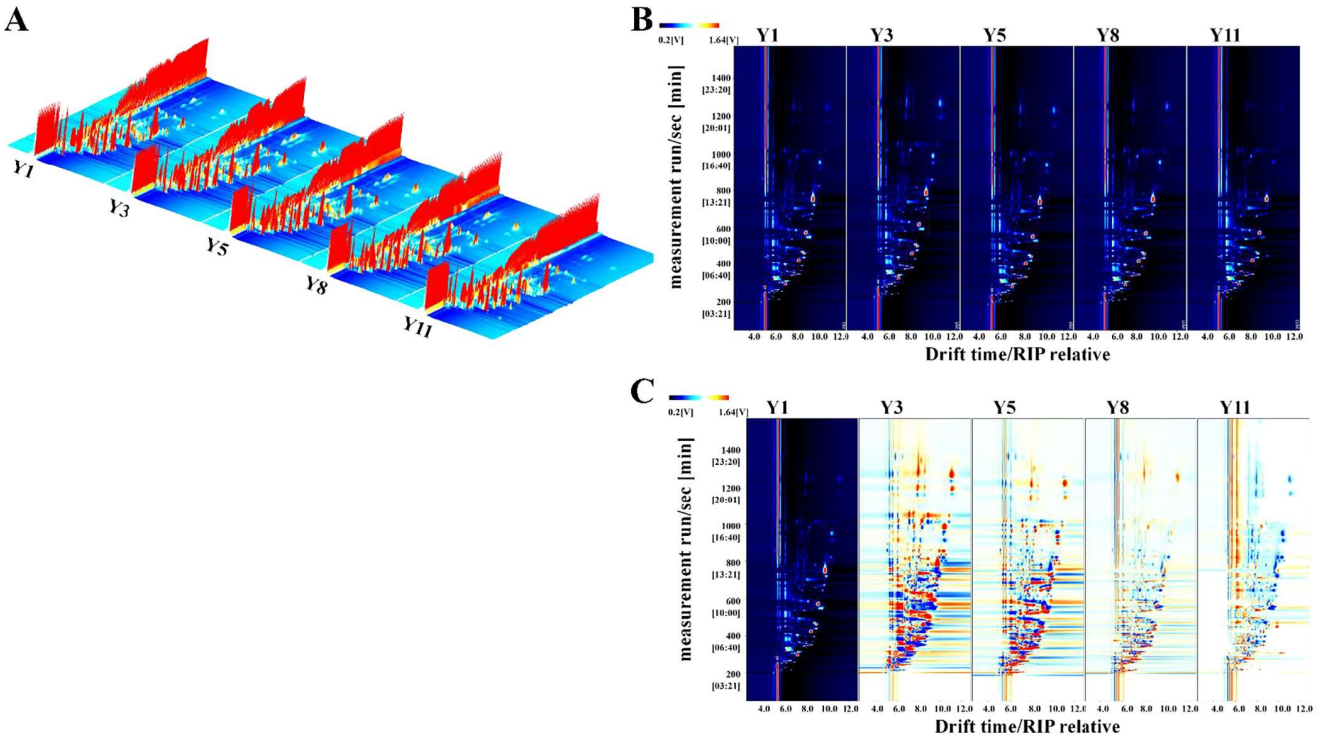


Figure 1. Volatile flavor compounds identified by HS-GC-IMS in different samples. (A) Comparison of 3D topographic plot. (B) 2D topographic plots. (C) Difference comparison plot (differences using Y1 as a reference).

volatile component that makes the greatest contribution to the overall flavor.

GC × GC-MS Analysis of Volatile Compounds. The volatile compounds in the samples were detected according to the previously,[10] a Shimadzu QP2020NX (Shimadzu Enterprise Management Co., Ltd.; Shanghai; China) was used for GC × GC−MS analysis. The one-dimensional column was the polar DB-WAX (30 m × 0.25 mm × 0.25 μm), and the two-dimensional column was the medium polar DB-17 ms (1.2 m × 0.18 mm × 0.18 μm). For the Baijiu samples (5 mL), 1.5 g of NaCl and 20 μL of the internal standard solution (2-octanol, with a concentration of 822 μg/mL) were added. The column carrier gas used for the column was high-purity helium (99.999%), with a flow rate of 1 mL/min. Regarding the oven temperature settings: initially, it was kept at 40 °C for 5 min; subsequently, the temperature was raised to 230 °C at a rate of 5 °C/min. The mass spectra were obtained under electron ionization mode. The ionization energy was set at 70 eV, and the scanning range was extended from m/z 25 to 550 amu. The ion source temperature was set to 230 °C. After extraction and processing of the original peaks, they were verified by comparing the mass spectra and retention indexes with those in the NIST17 (United States National Institute of Standards and Technology) database. The relative content of each compound was calculated by dividing the peak area of the separated compound by that of the internal standard.[11] The OAV method was employed to analyze the contribution of volatile components to the flavor. The calculation formula of the OAV is presented as follows

$$\mathrm{OAV}_x = \frac{C\%_x}{T}$$

OAV_x represents the odor activity value of the volatile component, while $C\%_x$ and T respectively denote the concentration and threshold of the volatile component.

Analysis of E-Nose. E-nose analysis was conducted with a PEN3 E-nose (Airsense Analytics GmbH, Schwerin, Germany), the procedure was founded on the method presented by Zhang et al.,[12] with some minor adjustments. Ten gas sensors made of metal oxide were utilized. Each of these sensors might exhibit distinct sensory characteristics, which were determined by its sensitivity to various compounds. Based on the performance of the E-nose, W1C, W5S, W3C, W6S, W5C, W1S, W1W, W2S, W2W, and W3S were successively sensitive to aromatic compounds; nitrogen oxides; ammonia and aromatic compounds; hydrogen; alkanes and aromatics; methane and a wide variety of compounds; sulfur compounds and terpenes; alcohols, aldehydes, and ketones; aromatics and organic sulfur compounds; long-chain alkanes.[13] The 0.1 mL sample and 1.49 mL of water were put into a 10 mL headspace vial. After that, the vial was covered with a cap and allowed to remain at room temperature for 30 min. Before testing, the equipment needs to be preheated for a minimum of 30 min. After each test, the mixture needs to be cleaned and standardized. The test parameters were set up in the following way: the temperature was kept at 20 ± 2 °C; the time required for sample preparation was 5 s. The duration of the measurement was 180 s, the measurement interval lasted for 1 s, the measurement duration was 1 s, the cleaning time of the sensor was 210 s, and the autozero time was set to 1 s; the flow rate of intake was set to 300 mL/min. Every sample was measured three times.

Statistical Analysis. All experiments were conducted three times, and the data were presented in the form of mean ± standard deviation (SD). To evaluate statistical differences ($P < 0.05$), analysis of variance (ANOVA) was conducted using SPSS 22.0. The difference comparison plot and fingerprint plot of volatile flavor compounds were created using the Gallery Plot plugin and Reporter plugin. OPLS-DA was carried out using the SIMCA (14.1, Umetrics, Umea, Sweden) software, a 200-times permutation test, and VIP analysis. Using the online tool of volatile compounds (https://www.omicshare.com) for principal component analysis (PCA) and drawing the VIP value with the content of point bar heat maps, E-nose response values were used to perform PCA and draw a radar map with Origin 2021.

RESULTS AND DISCUSSION

Identification of Volatile Aroma Compounds in Baijiu by GC-IMS. The qualitative analysis of volatile compounds was conducted by contrasting the retention times (Rt) and Dt of these volatile compounds. Additionally, it entailed making a

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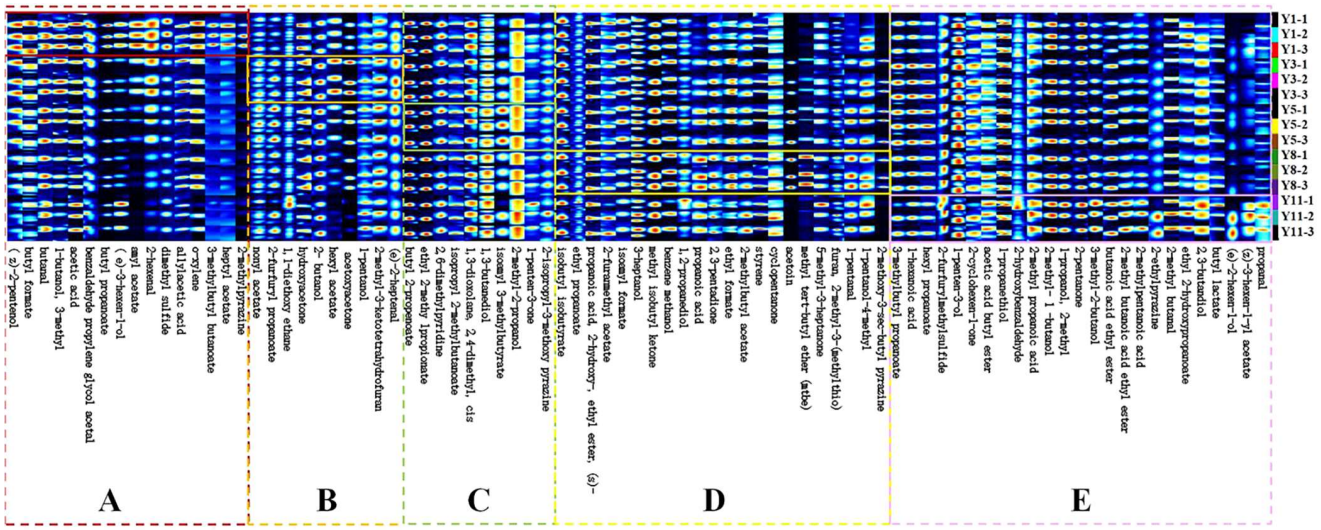


Figure 2. Fingerprint plot of volatile flavor compounds.

comparison between the NIST database incorporated into the GC-IMS Library Search and the Dt database. As shown in Figure 1A, the volatile compounds of NXX baijiu exhibited similar peaks in terms of signal distribution across different storage periods. Nevertheless, the intensities of the peak signals varied among different samples, suggesting that the content of volatile compounds varied among different samples. The two-dimensional topographic map (Figure 1B) showed that most volatile compounds were located at retention and Dt of 200−1000 s and 4−12 ms, respectively. To clearly distinguish the differences among the samples, we chose the spectrum of Y1 as the control, and a comparison diagram of the differences was obtained (Figure 1C). Compared with the control Y1, white indicated that the concentration of volatile compounds in the other samples was equal to that of Y1, red indicated that it was higher than that of Y1, and blue indicated that it was lower than that of Y1. Compared with Y1, the concentration of volatile compounds decreased with the increase of storage time during Rt and Dt of 1000−1400 s and 6−12 ms. In the Dt range of 5.5−6.5 ms, the concentration of volatile compounds rose along with the extension of storage time. The aforesaid results indicated that despite the general similarity in the spectra of the five samples, most volatile compounds exhibited notable differences in the location, number, and intensity of peaks.[14]

To visualize the variation patterns of individual volatile compounds within the sample, a fingerprint was created based on the detected volatile compounds. As depicted in Figure 2, each horizontal line represents a specific class of volatile compound within a single sample. Meanwhile, each vertical line indicated the variation of a specific volatile compound among multiple samples.[15] Each signal peak corresponded to a different compound, whose hue intensity was positively correlated with peak intensity and concentration. The fingerprint could be classified into five regions, namely, A, B, C, D, and E. In these regions, the concentration of volatile compounds reached its maximum in Y1, Y3, Y5, Y8, and Y11, respectively. In zone A, the concentrations of (*z*)-2-pentenol, butyl formate, 1-butanol, 3-methyl, benzaldehyde propylene glycol acetal, butyl propanoate, amyl acetate, 2-hexenal, allylacetic acid, and *o*-xylene gradually decreased with increasing years. In zone B, the concentrations of nonyl acetate, 2-furfuryl propanoate, hydroxyacetone, 2-butanol, hexyl acetate, acetoxyacetone, and (*e*)-2-heptenal first rose and then declined as the years went by. With the increase in years, the concentrations of isopropyl 2-methylbutanoate, 1,3-butanediol, isoamyl 3-methylbutyrate, and 2-isopropyl-3-methoxy pyrazine in zone C first rose and then declined. In zone D, the concentrations of 3-heptanol, methyl isobutyl ketone, propanoic acid, 2,3-pentadione, acetoin, methyl *tert*-butyl ether (mtbe), and 1-pentanal first rose and then declined as the number of years increased. The concentration of 1-propanethiol, 1-propanol, 2-methyl, butanoic acid ethyl ester, and 2-methyl butanoic acid ethyl ester in the E region increased gradually with the increase in years.

A total of 83 volatile compounds were identified in the 5 samples, including 28 esters, 17 alcohols, 10 ketones, 9 aldehydes, 6 acids, 4 pyrazines, 3 sulfides, and 6 others (Table 1). Esters (37.33−41.16%) and alcohols (22.78−28.05%) were the main volatile compounds. As shown in Figure 3A, the concentration of esters in Y5 was notably higher compared with that in other samples. Similarly, the ketone concentration in Y8 was notably higher than that in the remaining samples. According to the PCA score plot (Figure 3B), the principal components (PC), namely, PC1 and PC2, respectively, accounted for 32.4 and 20.2% of the total variance; the cumulative contribution of the first two principal components reached 52.6%, suggesting that they were sufficient to interpret the total variance within the data set. PC1 was positively correlated with Y8 and Y11, meanwhile, it had a negative correlation with Y1, Y3, and Y5. The distance between Y11 and the rest of the samples indicated that the aroma characteristics of Y11 differed significantly from those of other samples. In contrast, the distance between Y5 and Y8 showed that their aroma characteristics were similar. Based on GC-IMS data set, OPLS-DA was used to visualize data and perform quantitative analysis of the differences among samples (Figure 3C).[16] The various samples were distinctly separated, Y1 and Y3 were situated in the fourth quadrant, while the other samples were situated in different quadrants. This indicated that the aroma characteristics of compounds Y1 and Y3 were similar. The reliability of OPLS-DA was confirmed by

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Table 1. Volatile Flavor Compounds Identified by GC-IMS[a]

no.	flavor components	CAS#	formula	RI	retention time [s]	drift time [ms]	relative content (%)				
							Y1	Y3	Y5	Y8	Y11
	esters										
1	butyl lactate	C138227	C7H14O3	1013.50	340.86	1.27	0.47 ± 0.08a	0.49 ± 0.04a	0.24 ± 0.22b	0.46 ± 0.05a	0.52 ± 0.07a
2	ethyl 2-hydroxypropanoate	C97643	C5H10O3	1332.40	493.94	1.54	0.32 ± 0.03a	0.2 ± 0.08ab	0.11 ± 0.12b	0.26 ± 0.04a	0.32 ± 0.07a
3	2-furanmethyl acetate	C623176	C7H8O3	996.00	714.03	1.83	3.39 ± 0.61a	2.39 ± 1.01ab	1.39 ± 1.16b	2.9 ± 0.25a	2.54 ± 0.07ab
4	isoamyl 3-methylbutyrate	C659701	C10H20O2	1098.70	1275.25	2.04	0.95 ± 0.1b	2.32 ± 0.4a	1.24 ± 1.09ab	1.75 ± 0.02ab	1.45 ± 0.93ab
5	ethyl propanoate	C105373	C5H10O2	925.00	770.44	1.45	1.38 ± 0.05a	1.45 ± 0.29a	0.74 ± 0.57b	1.42 ± 0.03a	1.48 ± 0.2a
6	propanoic acid, 2-hydroxy-, ethyl ester, (s)-	C687478	C5H10O3	808.90	702.52	1.54	2.45 ± 0.11a	2.37 ± 0.09a	1.23 ± 1.01b	2.43 ± 0.07a	2.62 ± 0.25a
7	hexyl acetate	C142927	C8H16O2	1017.90	855.60	1.90	0.47 ± 0.02a	0.6 ± 0.1a	0.19 ± 0.16b	0.21 ± 0.01b	0.14 ± 0.07b
8	butyl formate	C592847	C5H10O2	823.67	519.63	1.21	0.95 ± 0.07a	0.97 ± 0.2a	0.4 ± 0.34b	0.78 ± 0.03ab	0.76 ± 0.22ab
9	butanoic acid ethyl ester	C105544	C6H12O2	1009.00	985.08	1.54	0.78 ± 0.04ab	1.01 ± 0.06a	0.52 ± 0.44b	1.06 ± 0.03a	1.08 ± 0.05a
10	acetic acid butyl ester	C123864	C6H12O2	1068.71	384.82	1.23	1.31 ± 0.13a	1.51 ± 0.21a	0.7 ± 0.58b	1.34 ± 0.02a	1.58 ± 0.26a
11	3-methylbutyl butanoate	C106274	C9H18O2	1056.10	434.60	1.41	0.08 ± 0.01a	0.04 ± 0.02b	0.01 ± 0.01b	0.03 ± 0.01b	0.04 ± 0.01b
12	2-methyl butanoic acid ethyl ester	C7452791	C7H14O2	855.30	662.40	1.64	1.02 ± 0.2a	0.78 ± 0.14ab	0.45 ± 0.36b	1 ± 0.15a	1.05 ± 0.1a
13	acetoxyacetone	C592201	C5H8O3	495.90	215.25	1.23	0.32 ± 0.03b	0.54 ± 0.02a	0.24 ± 0.19bc	0.39 ± 0.01b	0.15 ± 0.03c
14	(*Z*)-3-hexen-1-yl acetate	C3681718	C8H14O2	1309.90	288.49	1.04	0.11 ± 0.05ab	0.07 ± 0.03bc	0.03 ± 0.04c	0.05 ± 0bc	0.14 ± 0.02a
15	2-furfuryl propanoate	C623198	C8H10O3	1086.10	1249.96	1.46	1.35 ± 0.12a	2.39 ± 0.35a	1.25 ± 1.09a	1.95 ± 0.02a	1.63 ± 0.86a
16	2-methylbutyl acetate	C624419	C7H14O2	880.20	671.19	1.73	0.26 ± 0.01ab	0.22 ± 0.03ab	0.12 ± 0.1b	0.32 ± 0.13a	0.23 ± 0.04ab
17	3-methylbutyl propanoate	C105680	C8H16O2	974.80	954.19	1.93	2 ± 0.1ab	2.75 ± 0.12a	1.27 ± 1.07b	2.11 ± 0.02ab	1.69 ± 0.6ab
18	amyl acetate	C628637	C7H14O2	1160.73	645.70	1.76	0.38 ± 0.01a	0.27 ± 0.02b	0.1 ± 0.1c	0.17 ± 0.01c	0.12 ± 0.04c
19	butyl 2-propenoate	C141322	C7H12O2	896.60	616.07	1.69	7.57 ± 0.04a	6.93 ± 0.13a	3.4 ± 2.84b	6.37 ± 0.03a	6.49 ± 0.69a
20	butyl propanoate	C590012	C7H14O2	1119.08	260.61	1.29	0.39 ± 0.03bc	0.38 ± 0.11bc	0.27 ± 0.2c	0.62 ± 0.04a	0.51 ± 0.04ab
21	ethyl 2-methy lpropionate	C97621	C6H12O2	958.98	426.33	1.57	4.03 ± 0.05a	3.63 ± 0.07a	22.57 ± 35.13a	4.1 ± 0.02a	4.33 ± 0.06a
22	ethyl formate	C109944	C3H6O2	630.00	255.86	1.07	0.59 ± 0.06a	0.62 ± 0.04a	0.29 ± 0.27b	0.66 ± 0.03a	0.53 ± 0.03a
23	heptyl acetate	C112061	C9H18O2	1330.50	436.38	1.46	0.06 ± 0.01a	0.05 ± 0.02ab	0.01 ± 0.01c	0.03 ± 0bc	0.03 ± 0.01bc
24	hexyl propanoate	C2445763	C9H18O2	1310.67	1012.66	2.00	0.16 ± 0.02ab	0.26 ± 0.02a	0.13 ± 0.12b	0.25 ± 0a	0.23 ± 0.01ab
25	isoamyl formate	C110452	C6H12O2	769.20	782.08	1.64	0.43 ± 0.04a	0.46 ± 0.05a	0.22 ± 0.18b	0.35 ± 0.02ab	0.36 ± 0.02ab
26	isobutyl isobutyrate	C97858	C8H16O2	903.80	752.05	1.80	6.76 ± 0.08a	6.23 ± 0.21a	3.28 ± 2.76b	6.41 ± 0.08a	6.14 ± 0.57a
27	isopropyl 2-methylbutanoate	C66576714	C8H16O2	872.00	579.01	1.29	0.41 ± 0.05ab	0.46 ± 0.06a	0.25 ± 0.2b	0.49 ± 0.03a	0.46 ± 0.08a
28	nonyl acetate	C143135	C11H22O2	1325.40	1286.14	1.59	0.76 ± 0.02ab	0.99 ± 0.11a	0.5 ± 0.43b	0.86 ± 0.01ab	0.73 ± 0.13ab
	sum						39.16 ± 0.95a	40.39 ± 0.98a	41.16 ± 35.07a	38.76 ± 0.44a	37.33 ± 2.53a
	alcohols										
29	2,3-butandiol	C513859	C4H10O2	1526.23	667.13	1.37	0.87 ± 0.03a	0.78 ± 0.1a	0.42 ± 0.35b	0.87 ± 0.01a	0.97 ± 0.01a
30	1-propanol, 2-methyl	C78831	C4H10O	632.20	368.72	1.38	0.8 ± 0.07ab	0.91 ± 0.17a	0.47 ± 0.37b	1 ± 0.05a	1.1 ± 0.11a
31	2-methyl- 1 -butanol	C137326	C5H12O	735.70	378.41	1.46	1.4 ± 0.01a	1.16 ± 0.15a	0.61 ± 0.54b	1.18 ± 0.01a	1.28 ± 0.04a
32	1-butanol, 3-methyl	C123513	C5H12O	1208.07	363.13	1.49	0.99 ± 0.02a	0.82 ± 0.06ab	0.38 ± 0.34c	0.65 ± 0.02bc	0.63 ± 0.07bc
33	benzene methanol	C100516	C7H8O	1037.30	674.26	1.79	0.66 ± 0.02a	0.59 ± 0.11ab	0.34 ± 0.29b	0.74 ± 0.03a	0.58 ± 0.14ab
34	1-pentanol	C71410	C5H12O	1245.89	509.57	1.25	0.67 ± 0.02a	0.69 ± 0.04a	4.6 ± 7.16a	0.67 ± 0.02a	0.68 ± 0.06a
35	(*E*)-2-hexen-1-ol	C928950	C6H12O	1398.50	625.02	1.82	0.05 ± 0.03a	0.06 ± 0.02a	0.04 ± 0.03a	0.05 ± 0.03a	0.15 ± 0.11a
36	2-butanol	C78922	C4H10O	1026.00	1014.08	1.33	1.02 ± 0.01a	1.09 ± 0.02a	0.57 ± 0.48b	1.12 ± 0.02a	1.05 ± 0.12a
37	2-methyl-2-propanol	C75650	C4H10O	540.60	518.04	1.13	3.19 ± 0.09a	3.49 ± 0.27a	1.69 ± 1.47b	3.39 ± 0.05a	3.54 ± 0.29a

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Table 1. continued

no.	flavor components	CAS#	formula	RI	retention time [s]	drift time [ms]	relative content (%)				
							Y1	Y3	Y5	Y8	Y11
38	(*E*)-3-hexen-1-ol	C928972	C6H12O	1353.78	305.30	1.25	0.72 ± 0.16a	0.57 ± 0.05a	0.29 ± 0.24b	0.56 ± 0.03a	0.76 ± 0.06a
39	(*Z*)-2-pentenol	C1576950	C5H10O	747.40	398.96	1.62	2.9 ± 0.07a	2.55 ± 0.14a	12.96 ± 19.31a	2.29 ± 0.06a	2.23 ± 0.28a
40	1,2-propanediol	C57556	C3H8O2	1541.60	726.94	1.13	0.3 ± 0.04ab	0.34 ± 0.1a	0.15 ± 0.13b	0.36 ± 0.01a	0.36 ± 0.15a
41	1,3-butanediol	C107880	C4H10O2	1353.67	653.02	1.13	2.75 ± 0.04a	2.64 ± 0.27a	1.47 ± 1.23b	3.09 ± 0.03a	3.17 ± 0.49a
42	1-pentanol-4-methyl	C626891	C6H14O	843.70	712.62	1.32	0.44 ± 0.07a	0.41 ± 0.06ab	0.2 ± 0.18b	0.48 ± 0.06a	0.41 ± 0.16ab
43	1-penten-3-ol	C616251	C5H10O	656.50	334.00	1.34	3.42 ± 0.22a	2.81 ± 1.48a	1.73 ± 1.73a	3.61 ± 0.2a	3.36 ± 0.08a
44	3-heptanol	C589822	C7H16O	896.90	461.47	1.66	2.98 ± 0.02a	2.89 ± 0.23a	1.59 ± 1.3b	3.41 ± 0.06a	3.37 ± 0.67a
45	3-methyl-2-butanol	C598754	C5H12O	1020.60	538.11	1.45	1.11 ± 0.02ab	0.97 ± 0.09b	0.56 ± 0.47c	1.12 ± 0.01ab	1.41 ± 0.12a
	sum						24.28 ± 0.41a	22.78 ± 1.83a	28.05 ± 21.28a	24.6 ± 0.09a	25.05 ± 0.04a
	sulphides										
46	dimethyl sulfide	C75183	C2H6S	777.00	860.86	0.96	0.28 ± 0.01a	0.22 ± 0.02ab	0.11 ± 0.08c	0.18 ± 0.01bc	0.19 ± 0b
47	1-propanethiol	C107039	C3H8S	616.30	326.07	1.17	7.82 ± 0.03a	7.76 ± 0.14a	4.1 ± 3.29b	7.89 ± 0.11a	8.18 ± 0.7a
48	2-furfurylmethylsulfide	C1438911	C6H8OS	996.90	984.07	1.14	0.59 ± 0.02a	0.49 ± 0.1ab	0.28 ± 0.24b	0.61 ± 0.03a	0.67 ± 0.19a
	sum						8.69 ± 0.03a	8.46 ± 0.22a	4.49 ± 3.61b	8.68 ± 0.09a	9.04 ± 0.9a
	aldehydes										
49	1-pentanal	C110623	C5H10O	974.62	277.89	1.42	0.06 ± 0b	0.07 ± 0.03b	0.05 ± 0.04b	0.12 ± 0a	0.11 ± 0.01a
50	1,1-diethoxy ethane	C105577	C6H14O2	876.70	863.02	1.13	2.42 ± 0.14ab	2.34 ± 0.22ab	1.17 ± 0.98b	2.45 ± 0.02ab	2.84 ± 1.18a
51	2-methyl butanal	C96173	C5H10O	897.44	315.53	1.41	0.16 ± 0.01ab	0.11 ± 0.05b	0.08 ± 0.07b	0.16 ± 0.01ab	0.2 ± 0.02a
52	(*E*)-2-heptenal	C18829555	C7H12O	1302.81	784.44	1.25	0.24 ± 0.03a	0.26 ± 0.01a	0.14 ± 0.12b	0.24 ± 0.02a	0.27 ± 0.01a
53	1,3-dioxolane, 2,4-dimethyl, cis	C3390123	C5H10O2	700.40	404.37	1.33	1.37 ± 0.11a	1.4 ± 0.13a	0.67 ± 0.59b	1.28 ± 0.06a	1.41 ± 0.05a
54	2-hexenal	C505577	C6H10O	850.50	628.35	1.52	1.17 ± 0.14ab	1.53 ± 1.07a	0.43 ± 0.29b	0.59 ± 0.04ab	0.66 ± 0.14ab
55	2-hydroxybenzaldehyde	C90028	C7H6O2	1044.70	1052.36	1.13	1.74 ± 0.19a	1.52 ± 0.23ab	0.71 ± 0.6b	1.44 ± 0.01ab	1.78 ± 0.81a
56	butanal	C123728	C4H8O	839.00	306.09	1.29	0.95 ± 0.01a	0.92 ± 0.13a	0.43 ± 0.36b	0.84 ± 0.01a	0.93 ± 0.02a
57	propanal	C123386	C3H6O	786.52	214.70	1.05	0.13 ± 0.02b	0.16 ± 0.03ab	0.09 ± 0.09b	0.17 ± 0.02ab	0.23 ± 0.03a
	sum						8.24 ± 0.36a	8.3 ± 1.64a	3.78 ± 3.13b	7.29 ± 0.09a	8.42 ± 1.93a
	ketones										
58	2-pentanone	C107879	C5H10O	982.29	262.21	1.12	1.4 ± 0.06a	1.34 ± 0.02a	0.67 ± 0.6b	1.3 ± 0.01a	1.47 ± 0.17a
59	acetoin	C513860	C4H8O2	718.50	217.85	1.34	0.04 ± 0.01c	0.16 ± 0.01ab	0.11 ± 0.09bc	0.23 ± 0a	0.05 ± 0.01c
60	1-penten-3-one	C1629589	C5H8O	688.80	355.60	1.32	0.08 ± 0.02ab	0.07 ± 0.03ab	0.04 ± 0.04b	0.08 ± 0.01ab	0.1 ± 0.04a
61	2,3-pentadione	C600146	C5H8O2	1039.18	536.48	1.29	0.33 ± 0.02a	0.32 ± 0.04a	0.15 ± 0.14b	0.33 ± 0.01a	0.32 ± 0.02a
62	2-cyclohexen-1-one	C930687	C6H8O	936.00	960.96	1.41	1.29 ± 0.05a	1.41 ± 0.05a	0.69 ± 0.59b	1.29 ± 0.02a	1.16 ± 0.23ab
63	2-methyl-3-ketotetrahydrofuran	C3188009	C5H8O2	794.60	683.39	1.42	0.66 ± 0.06a	0.61 ± 0.09a	0.34 ± 0.27b	0.62 ± 0.03a	0.63 ± 0.05a
64	5-methyl-3-heptanone	C541855	C8H16O	942.00	359.46	1.28	0.32 ± 0.06ab	0.36 ± 0.08a	0.17 ± 0.15b	0.27 ± 0.05ab	0.37 ± 0.02a
65	cyclopentanone	C120923	C5H8O	791.00	787.35	1.34	1.15 ± 0.03a	1.18 ± 0.05a	0.62 ± 0.53b	1.22 ± 0.04a	1.22 ± 0.12a
66	hydroxyacetone	C116096	C3H6O2	714.10	243.99	1.23	1.29 ± 0.03a	1.39 ± 0.17a	0.69 ± 0.62b	1.46 ± 0.05a	1.14 ± 0.06ab
67	methyl isobutyl ketone	C108101	C6H12O	737.10	348.91	1.56	1.28 ± 0.03ab	1.31 ± 0.04ab	0.76 ± 0.63b	1.6 ± 0.01a	1.55 ± 0.19a
	sum						7.84 ± 0.13a	8.16 ± 0.18a	4.24 ± 3.65b	8.4 ± 0.14a	8 ± 0.42a
	acids										
68	acetic acid	C64197	C2H4O2	599.10	252.64	1.15	1.33 ± 0.03a	1.23 ± 0.17a	0.63 ± 0.6b	1.3 ± 0.04a	1.32 ± 0.04a
69	propanoic acid	C79094	C3H6O2	711.50	277.09	1.27	0.28 ± 0.03ab	0.27 ± 0.07ab	0.19 ± 0.15b	0.43 ± 0.02a	0.39 ± 0.05a
70	1-hexanoic acid	C142621	C6H12O2	1034.70	1015.08	1.63	0.73 ± 0.02ab	0.87 ± 0.05a	0.44 ± 0.37b	0.8 ± 0.01a	0.77 ± 0.04a

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Table 1. continued

no.	flavor components	CAS#	formula	RI	retention time [s]	drift time [ms]	relative content (%)				
							Y1	Y3	Y5	Y8	Y11
71	2-methyl propanoic acid	C79312	C4H8O2	790.30	494.04	1.38	2.31 ± 0.03a	2.28 ± 0.12a	13.13 ± 0.92a	2.18 ± 0.02a	2.43 ± 0.08a
72	2-methylpentanoic acid	C97610	C6H12O2	1034.10	541.94	1.59	0.69 ± 0.05a	0.5 ± 0.08a	0.28 ± 0.24b	0.59 ± 0.05a	0.67 ± 0.06a
73	allylacetic acid	C591800	C5H8O2	900.00	365.07	1.43	0.36 ± 0.06ab	0.38 ± 0.06a	0.18 ± 0.16b	0.31 ± 0.06ab	0.38 ± 0.1a
	sum						5.7 ± 0.03a	5.54 ± 0.1a	14.84 ± 2.43b	5.61 ± 0.03a	5.96 ± 0.13a
	pyrazines										
74	2-ethylpyrazine	C13925003	C6H8N2	1335.21	1342.60	1.53	0.16 ± 0.01a	0.29 ± 0.03a	0.16 ± 0.14a	0.26 ± 0.01a	0.33 ± 0.19a
75	2-isopropyl-3-methoxy pyrazine	C25773404	C8H12N2O	1410.83	713.27	1.25	0.31 ± 0.01a	0.28 ± 0.02a	0.15 ± 0.12b	0.33 ± 0.01a	0.31 ± 0.05a
76	2-methoxy-3-s-butyl pyrazine	C24168705	C9H14N2O	1423.40	219.58	1.28	0.04 ± 0.01b	0.11 ± 0a	0.07 ± 0.04b	0.11 ± 0a	0.05 ± 0b
77	2-methylpyrazine	C109080	C5H6N2	1258.57	355.70	1.09	0.08 ± 0.01a	0.04 ± 0.03b	0.02 ± 0.01c	0.03 ± 0bc	0.03 ± 0.01bc
	sum						0.6 ± 0.01ab	0.73 ± 0.05a	0.39 ± 0.31b	0.73 ± 0.02a	0.72 ± 0.14a
	others										
78	furan, 2-methyl-3-(methylthio)	C63012975	C6H8OS	1145.00	302.55	1.10	0.07 ± 0.01a	0.16 ± 0.14a	0.04 ± 0.04a	0.1 ± 0.01a	0.09 ± 0a
79	methyl tert-butyl ether (mtbe)	C1634044	C5H12O	641.70	198.63	1.13	0.55 ± 0.11b	0.84 ± 0.39ab	0.58 ± 0.46b	1.28 ± 0.04a	0.7 ± 0.08b
80	styrene	C100425	C8H8	876.70	344.44	1.51	0.57 ± 0.02a	0.5 ± 0.01a	0.26 ± 0.21b	0.51 ± 0.02a	0.52 ± 0.04a
81	*O*-xylene	C95476	C8H10	892.20	216.44	1.07	0.34 ± 0.01a	0.33 ± 0.04a	0.19 ± 0.13b	0.35 ± 0.01a	0.27 ± 0.05ab
82	benzaldehyde propylene glycol acetal	C2568254	C10H12O2	957.70	403.94	1.46	0.7 ± 0.02ab	0.92 ± 0.57a	0.33 ± 0.22b	0.55 ± 0.03ab	0.54 ± 0.03ab
83	2,6-dimethylpyridine	C108485	C7H9N	876.90	583.29	1.45	3.26 ± 0.2a	2.88 ± 0.62ab	1.64 ± 1.43b	3.14 ± 0.12a	3.34 ± 0.21a
	sum						5.49 ± 0.27a	5.64 ± 0.34a	3.03 ± 2.47b	5.94 ± 0.03a	5.46 ± 0.14a
	total sum						100.00	100.00	100.00	100.00	100.00

[a]Results represent the mean ± SD for three independent experiments. Mean values in the same row with the same letters are not significantly different at the 95% confidence level (Duncan's test).

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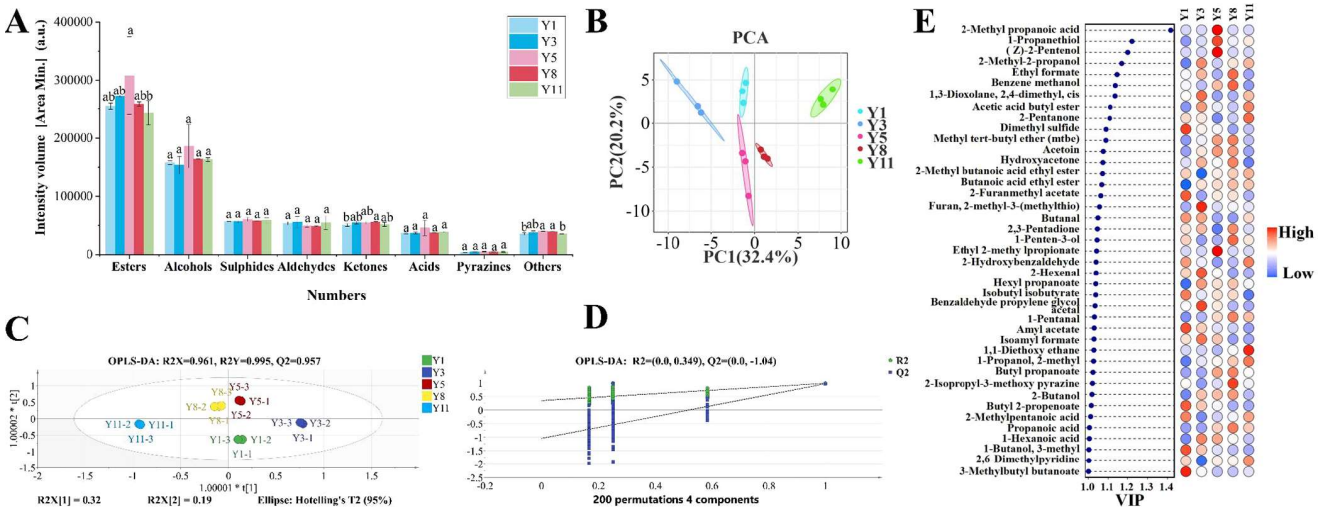


Figure 3. Results of obtained from GC-IMS. (A) The peak areas of each type of volatile flavor compounds. Values in the same pattern column with different superscript letters are significantly different ($p < 0.05$). (B) Score plots of PCA. (C) Score plots of OPLS-DA. (D) Cross-validation by a 200-times permutation test. (E) Volatiles with VIP ≥ 1.0 and $p < 0.05$.

Table 2. ROAV ≥ 1 Flavor Substance Number and Aroma Description[a]

						ROAV				
no.	flavor components	CAS	threshold (μg/L)	matrix	aroma description[g]	Y1	Y3	Y5	Y8	Y11
1	2-methyl butanoic acid ethyl ester	7452−79−1	18[b]	58% ethanol/ water solution	carrion, fruity, apple, wines, berry	100.00	100.00	100.00	100.00	100.00
2	1,1-diethoxy ethane	105−57−7	69[c]	46% ethanol/ water solution	fruity, cream	61.96	78.62	67.42	64.14	70.88
3	acetic acid butyl ester	123−86−4	58[b]	58% ethanol/ water solution	pear, fruity, banana	39.77	60.27	48.03	41.76	46.77
4	2-methyl propanoic acid	79−31−2	1580[d]	46% ethanol/ water solution	rancid, cheese	2.58	3.35	32.92	2.49	2.65
5	dimethyl sulfide	75−18−3	17[b]	58% ethanol/ water solution	cabbage, sulfur, corn, cooked onion	29.12	29.66	24.62	18.65	19.59
6	2-methyl butanal	96−17−3	16[c]	46% ethanol/ water solution	cocoa, almond	17.28	16.01	19.19	17.70	21.14
7	butanoic acid ethyl ester	105−54−4	140[f]	46% ethanol/ water solution	fruity, sweet	9.80	16.72	14.80	13.66	13.27
8	3-methylbutyl butanoate	106−27−4	20[b]	58% ethanol/ water solution	fruit, floral	6.74	4.86	2.93	2.74	3.21
9	2-methyl-2-propanol	75−65−0	2730[b]	58% ethanol/ water solution	alcoholic	2.06	2.97	2.46	2.24	2.23
10	butyl formate	592−84−7	800[b]	46% ethanol/ water solution	ripe fruits	2.10	2.82	1.97	1.76	1.63
11	1-pentanal	110−62−3	110[b]	58% ethanol/ water solution	special flavors	0.91	1.39	1.85	1.98	1.71
12	acetoin	513−86−0	259[b]	58% ethanol/ water solution	grease, creamy, sweet, fatty	0.28	1.47	1.72	1.59	0.30

[a]Odor threshold and aroma description references are as follows. [b]Reference 54. [c]Reference 55. [d]Reference 32. [e]Reference 56. [f]Reference 57. [g]The aroma descriptions for different compounds are derived from their threshold reference literature.

conducting 200 cross-validation tests on the model (Figure 3D). The goodness of fit of the model (R^2 = 0.728) and the goodness of prediction of the model (Q^2 = −1.2) were both less than 1, Moreover, the regression line of the model with a negative horizontal coordinate of Q^2 had a negative slope. This indicated that the model was not overfit, instead, it was stable and reliable. In addition, VIP scores were used to assess the significance of the compounds. A VIP value exceeding 1 was considered as a criterion for identifying potential marker metabolites.[17] In total, 41 volatile compounds with VIP > 1 were filtered out (Figure 3E), among which 2-methyl propanoic acid (VIP = 1.42) had the highest VIP value and contributed more to the differentiation of samples. Based on the established criteria, when VIP > 1 and $p \leq 0.05$,[18] 11 key aroma compounds were selected through screening (Figure 6A).

The relative content of 2-Methyl butanoic acid ethyl ester in each sample was substantial, while its threshold was relatively low. Thus, it was regarded as the primary flavor compound. The $ROAV_{stan}$ of it was set to 100, and the ROAV values of other volatile compounds are presented in Table 2. A total of 12 aromatic compounds whose ROAV was greater than or equal to 1 were detected. The ROAV (0.91) of 1-pentanal and the ROAV (0.28) of acetoin in Y1 were both less than 1,

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Table 3. Intensity of Volatile Flavor Compounds Identified by GC × GC-MS[a]

			average concentration (μg/mL)				
no.	flavor components	CAS	Y1	Y3	Y5	Y8	Y11
	esters						
1	ethyl hexanoate	123−66−0	5320.56 ± 47.54b	972.62 ± 17.26d	1104.58 ± 26.46c	6009.12 ± 29.01a	918.62 ± 3.73e
2	ethyl isovalerate	108−64−5	5.98 ± 1.16b	10.69 ± 9.76b	3842.22 ± 38.71a	23.36 ± 23.79b	13.13 ± 1.98b
3	ethyl butanoate	105−54−4	695.74 ± 32.93b	145.7 ± 7.17d	245.25 ± 28.18c	1202.85 ± 12.77a	262.58 ± 23.1c
4	ethyl acetate	141−78−6	1173.18 ± 11.17a	484.16 ± 9.79d	577.21 ± 21.11c	1054.47 ± 6.18b	129.49 ± 39.21e
5	ethyl caprylate	106−32−1	149.27 ± 48.98b	86.76 ± 46.87c	136.05 ± 10.32bc	572.73 ± 9.76a	71.71 ± 4.03d
6	ethyl valerate	539−82−2	439.66 ± 37.62a	114.58 ± 20.85c	80.4 ± 9.8c	393.75 ± 7.58b	82.3 ± 13.85c
7	ethyl heptanoate	106−30−9	162.42 ± 45.17b	67.59 ± 33.79 cd	92.79 ± 9.34c	359.81 ± 7.06a	41.72 ± 1.08d
8	ethyl lactate	97−64−3	301.95 ± 1.65a	5.42 ± 4.5d	58.95 ± 5.35c	107.81 ± 6.38b	17.43 ± 25.84d
9	ethyl myristate	124−06−1	7.1 ± 2.94bc	143.35 ± 3.36a	3.35 ± 1.42bc	7.4 ± 5.64b	0.9 ± 0.05c
10	ethyl 2-hydroxy-4-methyl valerate	10348−47−7	5.7 ± 2.58b	125.39 ± 1.28a	4.25 ± 0.71b	13.66 ± 12.7b	3.55 ± 0.03b
11	ethyl 4-methyl valerate	25415−67−2	8.56 ± 4.7b	124.14 ± 2.91a	7.47 ± 1.37b	11.92 ± 11.11b	3.11 ± 0.03b
12	octyl octanoate	2306−88−9	ND	120.81 ± 1.77a	1.44 ± 0.17b	2.34 ± 2.16b	ND
13	ethyl pentadecanoate	41114−00−5	0.81 ± 0.03b	120.73 ± 0.36a	ND	0.52 ± 0.05bc	0.29 ± 0.02c
14	ethyl nonanoate	123−29−5	ND	112.1 ± 0.71a	6.56 ± 0.4b	2.26 ± 0.05c	1.34 ± 0.05d
15	3-methylbutyl hexanoate	2198−61−0	16.37 ± 9.56b	104.74 ± 0.16a	16.43 ± 1.19b	34.11 ± 36.97b	14.8 ± 0.87b
16	ethyl linoleate	544−35−4	7.82 ± 2.76b	103.61 ± 2.15a	3.51 ± 1.82c	2.03 ± 0.83c	1.48 ± 1.38c
17	caproic acid propyl ester	626−77−7	4.67 ± 1.95bc	101.73 ± 2.56a	4.66 ± 0.75bc	8.35 ± 4.84b	1.35 ± 0.19c
18	methyl hexanoate	106−70−7	4.49 ± 0b	96.79 ± 0.08a	0.88 ± 0.05e	1.14 ± 0.03d	1.81 ± 0.19c
19	pentyl hexanoate	540−07−8	ND	94.42 ± 2.98a	3.48 ± 0.24b	ND	ND
20	hexyl hexanoate	6378−65−0	40.12 ± 24.28a	54.57 ± 37.62a	39.16 ± 5.36a	58.38 ± 55.01a	15.05 ± 1.57a
21	ethyl hexadecanoate	628−97−7	41.84 ± 10.73ab	47.81 ± 32.42a	14.02 ± 4.33ab	34.52 ± 32.12ab	3.56 ± 2.54b
22	butyl hexanoate	626−82−4	36.04 ± 21.36a	45.49 ± 38.46a	31.18 ± 6.24a	47.71 ± 45.08a	11.06 ± 0.48a
23	isobutyl hexanoate	105−79−3	ND	29.32 ± 48.04a	3.05 ± 0.07a	9.96 ± 5.96a	2.93 ± 0.12a
24	ethyl oleate	111−62−6	7.22 ± 2.55a	26.62 ± 40a	5.99 ± 1.4a	5.52 ± 4.17a	1.77 ± 1.19a
25	ethyl 3-hydroxybutyrate	5405−41−4	ND	7.35 ± 10.82a	ND	21.1 ± 16.8a	1.44 ± 0.08a
26	pentyl acetate	628−63−7	3.6 ± 1.99b	0.15 ± 0.02c	ND	20.24 ± 0.14a	ND
27	ethyl decanoate	110−38−3	10.44 ± 4.82a	14.36 ± 13.56a	9.38 ± 1.67a	19.31 ± 19.49a	7.47 ± 0.16a
28	butyl lactate	138−22−7	ND	18.03 ± 14.78	ND	ND	ND
29	ethyl laurate	106−33−2	ND	14.42 ± 18.54a	2.42 ± 0.74a	4.79 ± 4.53a	1.63 ± 0.55a
30	isopropyl hexanoate	2311−46−8	9.48 ± 0.34	ND	ND	ND	ND
31	butyl butanoate	109−21−7	3.79 ± 1.48a	9.03 ± 12.79a	2.35 ± 0.08a	7.09 ± 4.46a	0.95 ± 0.08a
32	ethyl 3-phenylpropanoate	2021−28−5	4.43 ± 1.73a	6.93 ± 5.98a	4.21 ± 1.04a	6.6 ± 5.27a	2.72 ± 0.12a
33	ethyl phenylacetate	101−97−3	6.62 ± 3.22a	5.24 ± 6.55a	1.9 ± 0.22a	3.05 ± 2.92a	2.52 ± 0.02a
34	ethyl formate	109−94−4	ND	ND	ND	6.27 ± 0.13	ND
35	methyl 4-methylpentanoate	2412−80−8	2.44 ± 1.43a	1.22 ± 0.61a	1.75 ± 0.02a	6.14 ± 3.51a	ND
36	octyl formate	112−32−3	ND	ND	2.12 ± 0.67a	6.08 ± 5.75a	1.97 ± 0.04a
37	isoamyl acetate	123−92−2	ND	5.9 ± 4.7a	ND	2.27 ± 0.7a	ND
38	heptyl hexanoate	6976−72−3	ND	ND	1.93 ± 0.56b	4.64 ± 0.03a	ND
39	9-hexadecenoic acid, ethyl ester	54546−22−4	3.85 ± 1.57a	ND	ND	ND	0.34 ± 0.02b
40	prop-2-enyl 2-ethyl butanoate	7493−69−8	ND	2.78 ± 4.33a	0.88 ± 0.16a	1.99 ± 0.74a	0.77 ± 0.28a
41	heptyl formate	112−23−2	ND	ND	2.55 ± 0.02a	ND	2.16 ± 0.03b
42	ethyl pyruvate	617−35−6	ND	2.19 ± 0.03	ND	ND	ND
43	elaidic acid ethyl ester	6114−18−7	ND	ND	ND	1.76 ± 0.02a	0.23 ± 0.06b
44	hexyl octanoate	1117−55−1	ND	1.64 ± 0.04	ND	ND	ND
45	ethyl stearate	111−61−5	1.26 ± 0.03	ND	ND	ND	ND
46	pentyl butyrate	540−18−1	ND	0.05 ± 0.02b	ND	ND	0.85 ± 0.03a
47	ethyl benzoate	93−89−0	ND	0.18 ± 0.03b	ND	ND	0.74 ± 0.12a
48	ethylidene diacetate	542−10−9	ND	0.73 ± 0.01	ND	ND	ND
49	butyl formate	592−84−7	ND	ND	ND	0.63 ± 0.02a	0.12 ± 0.01b
50	2-methylpentyl acetate	7789−99−3	ND	ND	ND	ND	0.25 ± 0.04
51	2-pentyl butyrate	60415−61−4	ND	ND	ND	ND	0.25 ± 0.05
52	furfuryl hexanoate	39252−02−3	ND	0.1 ± 0.02	ND	ND	ND
53	geranyl isovalerate	109−20−6	ND	ND	ND	ND	0.08 ± 0
54	isobutyl acetoacetate	7779−75−1	ND	ND	ND	ND	0.04 ± 0
	sum		8475.4 ± 241.2b	3429.46 ± 388.33d	6312.37 ± 105.66c	10075.69 ± 314.85a	1624.52 ± 20.89e
	alcohols						
55	acetal	105−57−7	212.95 ± 8.66e	472.55 ± 4.92c	676.65 ± 0.74b	2745.25 ± 42.33a	428.66 ± 5.63d

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Table 3. continued

no.	flavor components	CAS	average concentration (μg/mL)				
			Y1	Y3	Y5	Y8	Y11
56	pentanol	71−41−0	269.98 ± 13.74b	60.74 ± 31.92c	57.23 ± 7.01c	496.64 ± 1.95a	1.67 ± 1.47d
57	butanol	71−36−3	245.12 ± 20.77a	48.88 ± 26.51b	34.68 ± 4.9b	256.3 ± 37.57a	16.97 ± 0.09b
58	1-hexanol	111−27−3	129.01 ± 34.84b	45.58 ± 22.71c	37.5 ± 2.62c	245.76 ± 17.69a	21.15 ± 20.72c
59	phenethyl alcohol	60−12−8	ND	141.28 ± 1.31a	1.89 ± 0.89b	4.2 ± 3.87b	1.66 ± 0.66b
60	isobutanol	78−83−1	27.66 ± 15.37b	11.36 ± 6.21c	11.55 ± 1.15c	121.33 ± 3.92a	8.45 ± 6.81c
61	3-methyl-1-butanol	123−51−3	3.73 ± 0.05a	ND	0.59 ± 0.13a	31.74 ± 52.24a	34.52 ± 23.24a
62	1-octen-3-ol	3391−86−4	ND	ND	ND	ND	13.43 ± 0.29
63	3-methyl-2-butanol	598−75−4	5.82 ± 3.03a	1.78 ± 0.79a	2 ± 0.4a	8.43 ± 8.61a	3.39 ± 2.94a
64	propylene glycol	57−55−6	6.06 ± 3.87a	2.96 ± 0.02a	ND	0.63 ± 0.02a	ND
65	2-butanol	78−92−2	5.82 ± 0.06a	ND	ND	ND	1.61 ± 0.02b
66	1,2-hexanediol	6920−22−5	ND	ND	ND	2.56 ± 0.03	ND
67	cedrol	77−53−2	2.45 ± 0.1a	0.47 ± 0.29b	ND	ND	0.26 ± 0.02b
68	butane-2,3-diol	513−85−9	2.32 ± 0.05a	2.19 ± 1.62a	1.25 ± 0.39a	1.27 ± 0.02a	2.34 ± 2.29a
69	(−)-α-terpineol	10482−56−1	ND	ND	ND	ND	0.94 ± 0.02
70	2-heptanol	543−49−7	0.74 ± 0.41a	0.11 ± 0.01a	0.87 ± 0.7a	ND	0.13 ± 0.01a
71	linalool	78−70−6	ND	ND	ND	ND	0.61 ± 0.03
72	(*E*)-2-undecen-1-ol	75039−84−8	ND	ND	ND	ND	0.42 ± 0.03
73	1-octanol	111−87−5	ND	ND	ND	ND	0.4 ± 0.04
74	1-dodecanol	112−53−8	ND	ND	ND	ND	0.22 ± 0.01
75	1-nonanol	143−08−8	ND	0.16 ± 0.01	ND	ND	ND
76	2-hexanol	626−93−7	ND	ND	0.15 ± 0.02a	ND	0.14 ± 0.02a
77	(*S*)-(−)-2-methylbutanol	1565−80−6	ND	0.14 ± 0.01	ND	ND	ND
78	2-hexyl-1-decanol	2425−77−6	ND	ND	ND	ND	0.13 ± 0.03
79	2-nonanol	628−99−9	ND	ND	0.1 ± 0.01	ND	ND
	sum		911.65 ± 70.53b	788.2 ± 86.56c	824.46 ± 15.89bc	3914.11 ± 51.4a	537.09 ± 48.68d
	acids						
80	hexanoic acid	142−62−1	160.03 ± 35.47b	85.13 ± 13.98c	96.7 ± 22.62c	572.44 ± 23.32a	50.31 ± 43.11c
81	formic acid	64−18−6	59.4 ± 0.42b	48.64 ± 41.95b	ND	428.09 ± 1.86a	ND
82	isobutyric acid	79−31−2	4.17 ± 2.12b	141.71 ± 0.69a	0.66 ± 0.17b	4.13 ± 4.08b	0.37 ± 0.02b
83	3-methylvaleric acid	105−43−1	ND	126.52 ± 1.53a	1.85 ± 0.54b	5.34 ± 3.07b	5.58 ± 3.71b
84	pentanoic acid	109−52−4	7.6 ± 3.26b	124.88 ± 1.08a	3 ± 0.64b	8.1 ± 7.15b	2.12 ± 1.56b
85	octanoic acid	124−07−2	ND	114.61 ± 1.18a	15.08 ± 7.89bc	22.7 ± 2.14b	9.45 ± 0.74c
86	heptanoic acid	111−14−8	ND	53.31 ± 42.88a	6.1 ± 3.33a	4.57 ± 2.01a	3.89 ± 0.09a
87	butyric acid	107−92−6	41.62 ± 18.57a	17.58 ± 3.67ab	8.31 ± 1.42b	30.4 ± 22.48ab	36.16 ± 23.12ab
88	nonanoic acid	112−05−0	ND	ND	20.39 ± 9.65a	27.26 ± 0.02a	6 ± 0.25b
89	acetic acid	64−19−7	9.16 ± 4.02b	5.3 ± 0.71bc	15.08 ± 1.91a	9.66 ± 2.41b	3.11 ± 1.66c
90	2-nonenoic acid	3760−11−0	11.32 ± 0.7a	ND	1.16 ± 0.1b	1.57 ± 0.12b	ND
91	decanoic acid	334−48−5	ND	ND	4.09 ± 1.05a	5.22 ± 0.05a	1.53 ± 0.04b
92	undecanoic acid	112−37−8	ND	ND	ND	0.82 ± 0.05	ND
93	isocaproic acid	646−07−1	ND	0.15 ± 0.02	ND	ND	ND
	sum		293.3 ± 37.11c	717.84 ± 101.38b	172.4 ± 44.64d	1120.33 ± 28.73a	118.53 ± 38.33d
	aldehydes						
94	furfural	98−01−1	22.64 ± 15.45	ND	ND	ND	ND
95	acetaldehyde	75−07−0	1.65 ± 0.13b	1.8 ± 0.82b	ND	15.31 ± 0.97a	ND
96	phenylacetaldehyde	122−78−1	3.03 ± 1.72a	8.16 ± 5.97a	ND	ND	ND
97	benzaldehyde	100−52−7	ND	6.38 ± 3.83a	1.78 ± 0.1a	6.4 ± 4.3a	1.52 ± 0.22a
98	dodecanal	112−54−9	ND	ND	ND	ND	0.25 ± 0.03
99	heptanal	111−71−7	ND	ND	ND	ND	0.2 ± 0.02
100	4-undecanolide	104−67−6	ND	0.16 ± 0.03	ND	ND	ND
	sum		27.32 ± 17.23a	16.49 ± 9.19ab	1.78 ± 0.1b	21.71 ± 3.59a	1.97 ± 0.19b
	phenols						
101	*m*-cresol	108−39−4	3.12 ± 0.92b	138.28 ± 0.97a	ND	0.13 ± 0.03c	0.08 ± 0.01c
102	4-hydroxy-3-methoxystyrene	7786−61−0	ND	ND	ND	ND	6.48 ± 4.04
103	guaiacol	90−05−1	ND	ND	ND	ND	2.9 ± 0.05
104	phenol	108−95−2	2.8 ± 1.24a	ND	ND	ND	0.28 ± 0.03b
105	4-ethyl-2-methoxyphenol	2785−89−9	ND	ND	0.81 ± 0.05a	ND	0.49 ± 0.02b
106	2-ethylphenol	90−00−6	ND	ND	ND	ND	0.09 ± 0
107	eugenol	97−53−0	ND	ND	ND	ND	0.06 ± 0.01
	sum		5.92 ± 2.05c	138.28 ± 0.97a	0.81 ± 0.05d	0.13 ± 0.03d	10.38 ± 4.03b
	ketones						
108	2-octanone	111−13−7	ND	ND	ND	ND	63.59 ± 0.17

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Table 3. continued

			average concentration (μg/mL)				
no.	flavor components	CAS	Y1	Y3	Y5	Y8	Y11
109	butane-2,3-dione	431−03−8	ND	0.97 ± 0.04b	ND	5.73 ± 0.18a	ND
110	3-octanone	106−68−3	ND	ND	ND	ND	1.36 ± 0.02
111	1,1-diethoxynonane	54815−13−3	ND	ND	0.81 ± 0.02a	0.38 ± 0.05a	1.26 ± 0.18a
112	2-heptanone	110−43−0	ND	ND	0.5 ± 0.02a	ND	0.13 ± 0.01b
113	undecan-2-one	112−12−9	ND	ND	ND	ND	0.36 ± 0.02
114	2-pentadecanone	2345−28−0	ND	0.23 ± 0.02	ND	ND	ND
	sum		ND	1.2 ± 0.04c	1.3 ± 0.04c	6.11 ± 0.23b	66.71 ± 0.19a
	others						
115	limonene	138−86−3	ND	ND	ND	ND	3.14 ± 0.08
116	1,1-diethoxypentane	3658−79−5	ND	ND	1.39 ± 0.59a	ND	0.33 ± 0.02a
117	tetramethylpyrazine	1124−11−4	ND	ND	ND	ND	0.68 ± 0.06
118	2,2-difurfuryl ether	4437−22−3	0.36 ± 0.02	ND	ND	ND	ND
119	dimethyl trisulfide	3658−80−8	ND	ND	ND	ND	0.28 ± 0.02
	sum		0.36 ± 0.02c	ND	1.39 ± 0.59b	ND	4.43 ± 0.03a
	total sum		9713.95 ± 298.57b	5091.46 ± 540.1d	7314.51 ± 118.01c	15138.07 ± 381.82a	2363.63 ± 104.48e

[a]Results represent the mean ± SD for three independent experiments. Mean values in the same row with the same letters are not significantly different at the 95% confidence level (Duncan's test). ND = not detected.

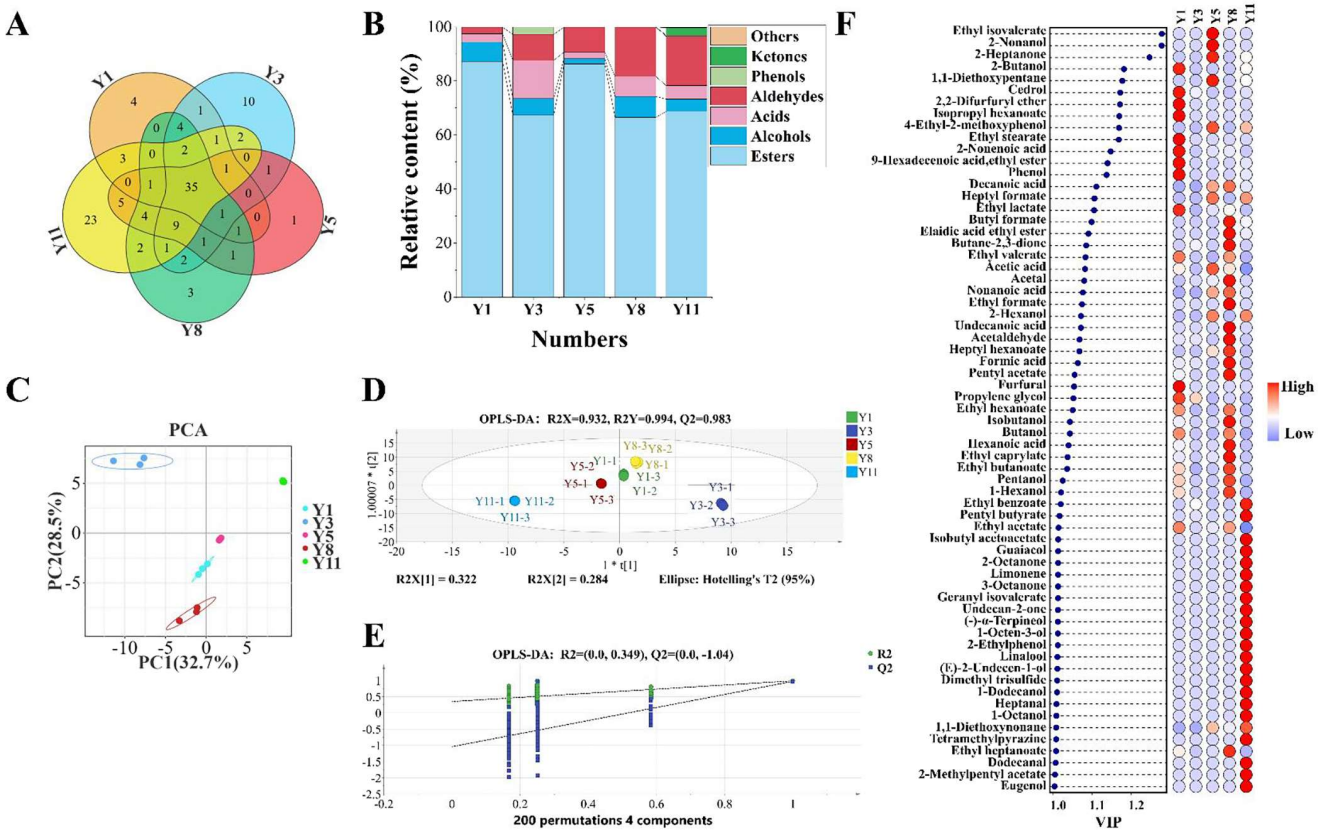


Figure 4. Results of obtained from GC × GC-MS. (A) Venn diagram of the number of volatile flavor compounds. (B) The relative content of each type of volatile flavor compounds. (C) Score plots of PCA. (D) Score plots of OPLS-DA. (E) Cross-validation by a 200-times permutation test. (F) Volatiles with VIP ≥ 1.0 and $p < 0.05$.

indicating that 1-pentanal and acetoin had modifying effects on the overall aroma of Y1. The ROAV of only acetoin in Y11 was 0.3, indicating that, in addition to acetoin, the remaining 11 volatile compounds were the key aroma compounds in Y11. 2-methyl-2-propanol had a high aroma intensity value, which had an important potential influence on the aroma characteristics of each sample. 3-methylbutyl butanoate exhibited a high ROAV value in Y1, resulting in a floral and fruity flavor in Y1. 1,1-diethoxy ethane, acetic acid butyl ester, dimethyl sulfide, butanoic acid ethyl ester, and butyl formate showed a higher ROAV value in Y3, significantly promoting the fruit, cabbage, and alcohol aroma of Y3. 2-methyl propanoic acid and acetoin exhibited high ROAV values in Y5, providing a cheesy and creamy flavor to Y5. 1-pentanal showed a high ROAV value in Y8, giving Y8 a special flavor. 2-methyl butanal showed a high ROAV value in Y11, making Y11 have cocoa and almond

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aroma. Among them, 1-pentanal could significantly distinguish the aroma difference between Y1 and the other samples. To improve screening accuracy, more stringent criteria (ROAV > 1.0, VIP > 1.0, and $p \leq 0.05$) were used to forecast potential key aroma compounds. The findings indicated that dimethyl sulfide and butanoic acid ethyl ester were identified as 2 potential key aroma compounds.

Identification of Volatile Aroma Compounds in Baijiu by GC × GC-MS. A total of 119 volatile compounds were detected via GC × GC-MS in 5 groups of samples. These included 54 esters, 25 alcohols, 14 acids, 7 aldehydes, 7 phenols, 7 ketones, and 5 other volatile compounds (Table 3). As shown in Figure 4A, 54,71,61,67, and 89 volatile compounds were detected in Y1, Y3, Y5, Y8, and Y11, respectively. Among these, 35 volatile compounds were detected in every sample, and Y11 had the most specific types of volatile compounds (23 kinds). The variety of volatile compounds increased as the storage time was prolonged; esters, alcohols, phenols, ketones, and others have the highest diversity in Y11. In Y8, esters, alcohols, and acids showed the highest contents, among these substances, esters particularly had the highest content in Y8 (10075.69 μg/mL), which resulted in a distinct fruity and floral aroma, enhancing the overall aroma.[19] According to research reports,[20] the activation energy for the protonation of the alkyl-oxygen in the ester was lower than that for the protonation of the hydroxyl-oxygen in the acid, which led to the fact that during the three-year storage of NXX baijiu, the hydrolysis of certain esters was more likely to occur than esterification. The ester concentration in Y3 (3429.46 μg/mL) was significantly lower than that in Y1 (8475.40 μg/mL), while the acid concentration in Y3 (717.84 μg/mL) was significantly higher than that in Y1 (293.30 μg/mL). During the storage of baijiu, if the concentration of a certain ester compound was relatively low, dissolved oxygen in the subsequent storage process would cause oxidation, increasing the concentration of the ester compound and reaching equilibrium in esterification.[21,22] The concentrations of ethyl butanoate (145.70−695.74 μg/mL) and ethyl hexanoate (972.62−5320.56 μg/mL) in Y3 were significantly lower than those in Y1, so the oxidation might increase the ester concentration in Y5 (6312.37 μg/mL) and Y8 (10075.69 μg/mL). Since the esterification reaction was reversible: if the ester concentration in baijiu was high, the esters would volatilize and hydrolyze, thereby reducing their concentration.[23] Compared with Y8, only the concentration of phenolic compounds (0.13−10.38 μg/mL) and pyrazine compounds (0−0.68 μg/mL) significantly increased in Y11, while the concentration of other types of compounds significantly decreased. This suggested that the significant reduction in the concentration of ester substances in baijiu during the 8−11 years of storage time might mainly be due to volatilization. Esters were the main volatile compounds, accounting for the total volatile compounds (66.56−87.25%), and alcohols were the second richest category, accounting for the total volatile compounds (9.38−25.86%), followed by acids (2.36−14.10%), aldehydes (0.08−0.32%), phenols (0.01−2.72%), ketones (0−2.83%), and others (0−0.19%) (Figure 4B).

Esters are important aroma compounds in baijiu, they are mainly generated through the esterification of acid and alcohol[24] as well as by esterification of yeast,[25] with a typical fruity, floral, and sweet taste.[26] Ethyl hexanoate was the highest content of straight chain ester in the sample (918.62−6009.12 μg/mL). The content of short-chain ester (9928.22 μg/mL) and medium-chain ester (87.11 μg/mL) in Y8 was the highest, and the total amount of medium-chain ester and long-chain ester in all samples showed the same change trend during storage. Ethyl ester was the main long linear chain ester, and its content fluctuated greatly in each sample (8.57−442.13 μg/mL). This concentration fluctuation might be due to the complex interaction between the formation and degradation process, and the increase was stimulated by the esterification reaction at the initial stage of storage,[27] the total content of long-chain fatty acid ethyl ester dropped remarkably as the storage time increased.[28] A notable positive correlation existed between the concentration of long-chain fatty acid ethyl ester and both the taste and the sweetness of the fruit. Moreover, a higher concentration could lead to a reduction of odor, irritation, and spiciness, indicating that Y3 had a better flavor. Hexyl hexanoate, ethyl isovalerate, and ethyl hexanoate were the main ingredients affecting the sweetness of baijiu,[29] the total content of the three samples reached the highest level in Y8, which indicated that its sweetness may be stronger than that of other samples. As the storage time increased, the total ester content in the sample presented a tendency of increasing significantly at first and then decreasing significantly, indicating that the equilibrium transformations of alcohol, acid, and ester in baijiu were likely the result of chemical reactions like oxidation, esterification, hydrolysis, and Maillard reaction that occurred during this period, which led to the unstable change of some ester compounds during long-term storage,[30] the hydrolysis of esters and the decomposition products of macromolecules affected the appearance of the baijiu,[31] and the presence of esters made the baijiu taste softer, sweeter, and more pure. Storage time led to dynamic alterations in the content and types of ester compounds in NXX baijiu and also led to differences in baijiu style characteristics. Compared with previous studies,[32,33] in all samples of this study, ethyl phenylacetate was detected, the concentration varied from 1.90 μg/mL to 6.62 μg/mL, which were similar to those in strong-flavored baijiu (7.52 μg/mL) and northern strong-aroma type baijiu (2.74−3.94 μg/mL).

Alcohols are the main source of mellow sweetness and aroma and also the precursor of esters in baijiu. It is primarily generated by microorganisms via sugar metabolism and amino acid metabolism.[34] Their existence will remarkably influence the flavor and contribute to the smoothness and overall quality of baijiu.[35] Alcohols are typically classified into two major groups: ethanol and other higher alcohols; ethanol usually has a relatively high concentration and is generally considered to be the main flavor substance rather than the aroma substance. Therefore, the influence of alcohol compounds on baijiu aroma is strongly associated with the existence of higher alcohols. When the proportion of higher alcohols is properly balanced, they will bring a sweet, beautiful, and mellow feeling to the baijiu and make the baijiu soft and mellow; however, an excessive amount of higher alcohols will produce a greasy and unpleasant odor and cause dizziness. The content and proportion of higher alcohols in the baijiu determine its flavor and feeling after drinking. Acetal, pentanol, butanol, 1-hexanol, and other high alcohols, as the main aroma substances in NXX baijiu, giving baijiu a fruity aroma, alcohol aroma, and nut aroma, which play a part in making the baijiu body full and rounded.[36] Oct-1-en-3-ol, (−)-α-terpineol, linalool, (*E*)-2-undecen-1-ol, 1-octanol, 1-dodecanol, and 2-hexyl-1-decanol contents increased gradually. The contents of acetal, 1,2-hexanediol, 1-nonanol, (*S*)-(−)-2-methylbutanol, and 2-non-

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anol increased first and then decreased, 1-nonanol and (*S*)-(−)-2-methylbutanol were only detected in Y3, 2-nonanol was only detected in Y5, 1,2-hexanediol was only detected in Y8, and the content of acetal was the highest in Y8. The main fusel oils detected were pentanol, butanol, isobutanol, 3-methyl-1-butanol, and 1-octanol, giving the fullness of baijiu taste and aroma, but slower in the decomposition rate of oxidation in the human body could cause neurological congestion and dry mouth, the total fusel oil content reached the maximum in Y8 and the minimum in Y11. Compared with previous studies,[37] it was found that the concentration of 3-methyl-1-butanol in aged NXX baijiu (3.73−34.52 μg/mL) were significantly lower than that in freshly distilled baijiu (79.1−372.00 μg/mL). These results indicated that proper control of the storage time can improve the comfort level after drinking and reduce the dryness and spiciness of NXX baijiu.

Acids are of great significance in shaping the flavor characteristics of baijiu, contributing to taste harmony, aroma stabilization, and enhancing color stability as well as oxidation resistance.[38] A deficiency in acidity within baijiu can lead to a diminished flavor experience. Conversely, excessive acid content may result in undesirable odors such as sweat and other unpleasant scents.[39] Beyond their contribution to flavor, acidic compounds also exhibit probiotic properties that support antioxidant activity and intestinal health maintenance.[40] Therefore, judiciously increasing the concentration of acidic substances within the baijiu matrix not only enhances its flavor but also imparts health benefits. In our analysis, volatile organic acids were found to be the most abundant in Y8 before experiencing a sharp decline. Among all samples analyzed, hexanoic acid exhibited the highest concentration and was recognized as one of the primary compounds responsible for baijiu aroma; it was characterized by descriptors such as "sweaty," "oily," 'tacky," and "sour".[41] At low concentrations, hexanoic acid contributed positively to fruity, floral, and sweet notes; however, it diminished the mellow and grainy aromas. At medium concentrations of hexanoic acid, there was a reduction in fruitiness, along with sweet and floral characteristics; nevertheless, this level yielded an overall higher aroma score. In contrast, high concentrations of hexanoic acid significantly enhanced the cellar-like aroma while markedly reducing floral sweetness and fruity attributes. This increase led to heightening sourness alongside pronounced sweat/cheese notes, resulting in an overall less harmonious aromatic profile. In conclusion, variations in the hexanoic acid content substantially influence the overall harmony of aromatic contours across different samples. Major short-chain fatty acids in the sample were hexanoic, formic, and isobutyric acids, with the highest total content in Y8, which can reduce the irritating odor and bitter taste of baijiu and enhance its sweetness. It also alleviated alcoholic liver disease and improved intestinal barrier integrity and immune function.[42] The amounts of heptanoic acid and isocaproic acid initially increased and then declined, with the highest concentrations observed in the Y3. In contrast, the level of butyric acid initially decreased and then increased and reached the minimum in Y5. The content of nonanoic acid and decanoic acid followed a similar trend by first increasing and subsequently decreasing; notably, their highest concentration was found in the Y8.

The balance between aldehydes and their acetals is essential for the aroma profile of aged distilled beverages. Aldehydes are characterized by fresh and spicy notes, while acetals contribute pleasant fruity undertones that enhance a ripening aroma.[43] With its low boiling point and high volatility, furfural could test the age and authenticity of the distilled beverage, imparting a toasty aroma and sweetness to the baijiu, and its content decreased significantly with longer storage time, which was consistent with previous studies,[6] because it was reduced to furfuryl alcohol and then reacted with hexanoic acid to produce furfuryl hexanoate.[44] The content of aldehyde volatile compounds was generally low and decreased significantly with the extension of storage time, which may be caused by gradual oxidation during storage. The content of ketone volatile compounds rose gradually during storage, and 2-octanone was the main ketone volatile compound, releasing "floral", "fruity", and "herbal" aromas. 4-Ethyl-2-methoxyphenol (0−0.81 μg/mL) was animal odor, stable odor, and had a detrimental effect on the overall flavor of the baijiu. Dimethyl trisulfide (0−0.28 μg/mL), the only detected sulfide, was the highest in 11Y and was thought to be the main compound of pickle flavor in some white spirits, presenting sulfur, cabbage, onion, and pickles flavors.[45]

As can be seen from the PCA score plot (Figure 4C), PC1 and PC2 respectively accounted for 32.7 and 28.5% of the total variance, and the combined contribution of the first two principal components reached 61.2%, suggesting that they were adequate to account for all of the variance within the data set. There was a positive correlation between PC1 and Y5 as well as Y11, while there was a negative correlation between PC1 and Y3 and also between PC1 and Y8. The distance between Y3 and the rest of the samples indicated that the flavor characteristics of Y3 differed significantly from those of the other samples. In contrast, the short distances among Y1, Y5, and Y8 indicated that their aroma characteristics were similar. Based on the GC × GC-MS dataset, the OPLS-DA method was used to evaluate the flavor differences among different samples. In the sample grouping process, the values of R^2X and R^2Y were 0.932 and 0.994, respectively, and as clearly presented in Figure 4D, there were notable differences between various samples. Among them, Y1 and Y8 were situated in the first quadrant, while the remaining samples were positioned in different quadrants. This clear separation between samples of different years highlighted the large influence of storage time on flavor characteristics, indicating that the flavor characteristics of baijiu with different storage times varied greatly. To investigate these differences thoroughly, we used a rank test (n = 200) to identify markers of characteristic differences among baijiu samples and assess whether the discriminant model showed signs of proper fitting. As shown in Figure 4E, the regression line of Q^2 intersected the vertical axis at a point below 0 (left). All of the Q^2 and R^2 values here were less than the origin values on the right. Specifically, the R^2Y intercept was about 0.349, while the Q^2 intercept was about −1.04, which was consistent with the recommended threshold for a well-fitted model,[46] thus confirming that the model was both robust and valid. A total of 65 potential biomarker metabolites with VIP > 1 were screened out using variable importance (VIP) to evaluate the significance of compounds (Figure 4F). Subsequently, 16 key aroma compounds were ultimately selected based on the criteria of VIP > 1 and $p \leq 0.05$ (Figure 6B).

The impact of aroma components on the overall flavor is affected not merely by their concentrations but also by the olfactory threshold of consumers.[47] This threshold exhibits significant variability in sensitivity toward different volatile compounds, which can be attributed to variations in chemical

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Table 4. OAV ≥ 1 Flavor Substance Number and Aroma Description[a]

no.	flavor components	CAS	threshold (μg/L)	matrix	aroma description[o]	OAV				
						Y1	Y3	Y5	Y8	Y11
1	ethyl isovalerate	108−64−5	7[b]	46% ethanol/ water solution	apple	854.22	1527.56	548888.80	3337.10	1875.00
2	ethyl hexanoate	123−66−0	55[b]	46% ethanol/ water solution	fruity	96737.42	17683.91	20083.21	109256.75	16702.20
3	ethyl caprylate	106−32−1	13[c]	46% ethanol/ water solution	fruit, sweet, floral	11481.94	6673.86	10465.28	44056.06	5516.40
4	acetal	105−57−7	69[d]	46% ethanol/ water solution	fruity, cream	3086.21	6848.55	9806.52	39786.26	6212.47
5	ethyl acetate	141−78−6	30[b]	46% ethanol/ water solution	pineapple	39106.11	16138.56	19240.46	35149.02	4316.31
6	ethyl valerate	539−82−2	27[c]	46% ethanol/ water solution	fruit, sweet, floral	16283.56	4243.83	2977.83	14583.34	3048.01
7	ethyl butanoate	105−54−4	140[e]	46% ethanol/ water solution	fruity, sweet	4969.57	1040.70	1751.78	8591.75	1875.57
8	1-octen-3-ol	3391−86−4	6[f]	52% ethanol/ water solution	mushroom	-	-	-	-	2237.60
9	methyl hexanoate	106−70−7	55[g]	65% ethanol/ water solution	fruity, pineapple	81.63	1759.82	15.98	20.70	32.90
10	dimethyl trisulfide	3658−80−8	0.36[e]	46% ethanol/ water solution	cabbage	-	-	-	-	775.06
11	2-octanone	111−13−7	190[h]	water	dairy, cheese,	-	-	-	-	334.68
12	pentanoic acid	109−52−4	389[h]	46% ethanol/ water solution	sweaty	19.54	321.03	7.71	20.83	5.46
13	ethyl 4-methyl valerate	25415−67−2	389[i]	58% ethanol/ water solution	fruit	21.99	319.13	19.21	30.65	7.99
14	octanoic acid	124−07−2	500[h]	46% ethanol/ water solution	sweat, cheese	ND	229.23	30.15	45.41	18.89
15	hexanoic acid	142−62−1	2520[b]	46% ethanol/ water solution	sweaty, rancid	63.50	33.78	38.37	227.16	19.97
16	guaiacol	90−05−1	13[j]	46% ethanol/ water solution	scented oil	-	-	-	-	223.41
17	isoamyl acetate	123−92−2	30[g]	65% ethanol/ water solution	fruity	-	196.66		75.76	-
18	furfural	98−01−1	122[b]	46% ethanol/ water solution	baked	185.58	-	-	-	-
19	ethyl linoleate	544−35−4	1000[c]	53% ethanol/ water solution	fruit, floral	7.82	103.61	3.51	2.03	1.48
20	ethyl 2-hydroxy-4-methyl valerate	10348−47−7	1268[k]	58% ethanol/ water solution	fruity, blackberry, freshness	4.49	98.89	3.35	10.77	2.80
21	butanol	71−36−3	2730[l]	46% ethanol/ water solution	fruity	89.79	17.90	12.70	93.88	6.22
22	4-hydroxy-3-methoxystyrene	7786−61−0	70[c]	53% ethanol/ water solution	smoky	-	-	-	-	92.63
23	isobutyric acid	79−31−2	1580[h]	46% ethanol/ water solution	rancid, cheese	2.64	89.69	0.42	2.62	0.23
24	butyl butanoate	109−21−7	110[k]	58% ethanol/ water solution	fruity, pineapple	34.48	82.13	21.38	64.49	8.68
25	3-methylbutyl hexanoate	2198−61−0	1400[l]	46% ethanol/ water solution	apple, fruity	11.70	74.82	11.74	24.37	10.57
26	heptanal	111−71−7	3[l]	46% ethanol/ water solution	aldehydic	-	-	-	-	66.02
27	1-hexanol	111−27−3	4000[e]	46% ethanol/ water solution	nutty	32.25	11.40	9.37	61.44	5.29
28	undecan-2-one	112−12−9	6[c]	53% ethanol/ water solution	fruit, wax	-	-	-	-	60.07
29	ethyl 3-phenylpropanoate	2021−28−5	125[e]	46% ethanol/ water solution	dried floral	35.43	55.48	33.70	52.77	21.78
30	butyric acid	107−92−6	964[b]	46% ethanol/ water solution	sweaty, rancid	43.18	18.24	8.62	31.54	37.51
31	ethyl laurate	106−33−2	400[c]	53% ethanol/ water solution	fruit, sweet	-	36.04	6.04	11.97	4.08
32	ethyl nonanoate	123−29−5	3150[m]	46% ethanol/ water solution	sweet fruity	-	35.59	2.08	0.72	0.42
33	phenylacetaldehyde	122−78−1	262[b]	46% ethanol/ water solution	honey, floral	11.58	31.13	-	-	-
34	hexyl hexanoate	6378−65−0	1890[n]	46% ethanol/ water solution	fruity	21.23	28.88	20.72	30.89	7.96
35	ethyl heptanoate	106−30−9	13,200[l]	46% ethanol/ water solution	fruity	12.30	5.12	7.03	27.26	3.16

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Table 4. continued

no.	flavor components	CAS	threshold (μg/L)	matrix	aroma description[o]	OAV				
						Y1	Y3	Y5	Y8	Y11
36	linalool	78−70−6	25[k]	58% ethanol/ water solution	citrus, orange, lemon, floral	-	-	-	-	24.36
37	3-octanone	106−68−3	70[k]	58% ethanol/ water solution	fumes, aldehyde	-	-	-	-	19.43
38	ethyl decanoate	110−38−3	1120[l]	46% ethanol/ water solution	fruity, grape	9.32	12.82	8.37	17.24	6.67
39	ethyl phenylacetate	101−97−3	407[b]	46% ethanol/ water solution	floral, sweet	16.27	12.87	4.66	7.49	6.20
40	pentanol	71−41−0	37,370[g]	65% ethanol/ water solution	fruity, alcoholic	7.22	1.63	1.53	13.29	0.04
41	acetaldehyde	75−07−0	1200[b]	46% ethanol/ water solution	fruity, grassy	1.37	1.50		12.76	-
42	butyl hexanoate	626−82−4	5250[f]	46% ethanol/ water solution	fermented fruit	6.87	8.66	5.94	9.09	2.11
43	caproic acid propyl ester	626−77−7	12,800[k]	58% ethanol/ water solution	pineapple, sweet, fruity	0.36	7.95	0.36	0.65	0.11
44	nonanoic acid	112−05−0	3559[h]	46% ethanol/ water solution	heavy acid	-	-	5.73	7.66	1.69
45	pentyl hexanoate	540−07−8	13,802[k]	58% ethanol/ water solution	fruity	-	6.84	0.25	-	-
46	4-ethyl-2-methoxyphenol	2785−89−9	123[i]	20% ethanol/ water solution	spice	-	-	6.57	-	3.94
47	benzaldehyde	100−52−7	990[c]	46% ethanol/ water solution	benzaldehyde	-	6.45	1.80	6.46	1.54
48	isobutyl hexanoate	105−79−3	5250[f]	53% ethanol/ water solution	sweet fruity rose	-	5.58	0.58	1.90	0.56
49	heptanoic acid	111−14−8	13,800[n]	46% ethanol/ water solution	cheesy, sweaty	-	3.86	0.44	0.33	0.28
50	2-heptanone	110−43−0	140[b]	46% ethanol/ water solution	fruity	-	-	3.55	-	0.96
51	phenethyl alcohol	60−12−8	40,927[f]	46% ethanol/ water solution	floral	-	3.45	0.05	0.10	0.04
52	1-nonanol	143−08−8	50[c]	53% ethanol/ water solution	floral, fat	-	3.27	-	-	-
53	ethyl myristate	124−06−1	46,607[f]	46% ethanol/ water solution	medicinal liquid	0.15	3.08	0.07	0.16	0.02
54	eugenol	97−53−0	21[k]	58% ethanol/ water solution	clove, woody	-	-	-	-	2.88
55	ethyl lactate	97−64−3	128,000[i]	46% ethanol/ water solution	mushroom	2.36	0.04	0.46	0.84	0.14
56	2-butanol	78−92−2	3300[l]	46% ethanol/ water solution	sweet, apricot	1.77	-	-	-	0.49
57	2-nonanol	628−99−9	70[k]	58% ethanol/ water solution	floral, fruity	-	-	1.45	-	-
58	ethyl hexadecanoate	628−97−7	39,299[k]	58% ethanol/ water solution	wax, grease, frankincense	1.06	1.22	0.36	0.88	0.09
59	isocaproic acid	646−07−1	144[c]	53% ethanol/ water solution	fat	-	1.06	-	-	-

[a]-, not detected. Odor threshold and aroma description references are as follows. [b]Reference 58. [c]Reference 59. [d]Reference 55. [e]Reference 57. [f]Reference 60. [g]Reference 61. [h]Reference 32. [i]Reference 41. [j]Reference 62. [k]Reference 54. [l]Reference 56. [m]Reference 37. [n]Reference 33. [o]The aroma descriptions for different compounds are derived from their threshold reference literature.

composition, molecular structure, and the specific degree of binding of these compounds is directed toward the olfactory receptor cells within the nasal cavity. The "odor threshold" is generally defined as the lowest concentration at which an individual can detect a volatile compound.[48] Consequently, the odor activity value (OAV) serves as a method for assessing two factors essential for identifying key odors in baijiu. This included estimating the OAV using previously established thresholds for aroma active ingredients and calculating the OAV of the compound by dividing the concentration of each target compound by its respective odor threshold (Table 4). Among the 5 samples analyzed, 59 aromatic compounds with an OAV of ≥1 were recognized. Among Y1, Y3, Y5, Y8, and Y11, 32, 42, 33, 33, and 39 compounds had an OAV ≥ 1, respectively, and 24 aromatic compounds with an OAV ≥ 1 were found in all samples.

These compounds, which are primarily formed during the processes of fermentation and aging, have a low aroma threshold, and they endow a strong fruity and floral flavor.[3] Ethyl isovalerate (apple) had the highest OAV in Y5 (548888.80), which was consistent with its high aroma intensity and concentration, indicating that it may be a key aromatic component of sample Y5, endowing Y5 with a rich apple flavor. Similarly, ethyl hexanoate had the highest OAV in Y8 (109256.75) and was the highest volatile compound in Y1, Y3, and Y11. Its aroma intensity and concentration in each sample were high, indicating that Y8 had the strongest fruit flavor; however, due to its very low threshold, it had a high

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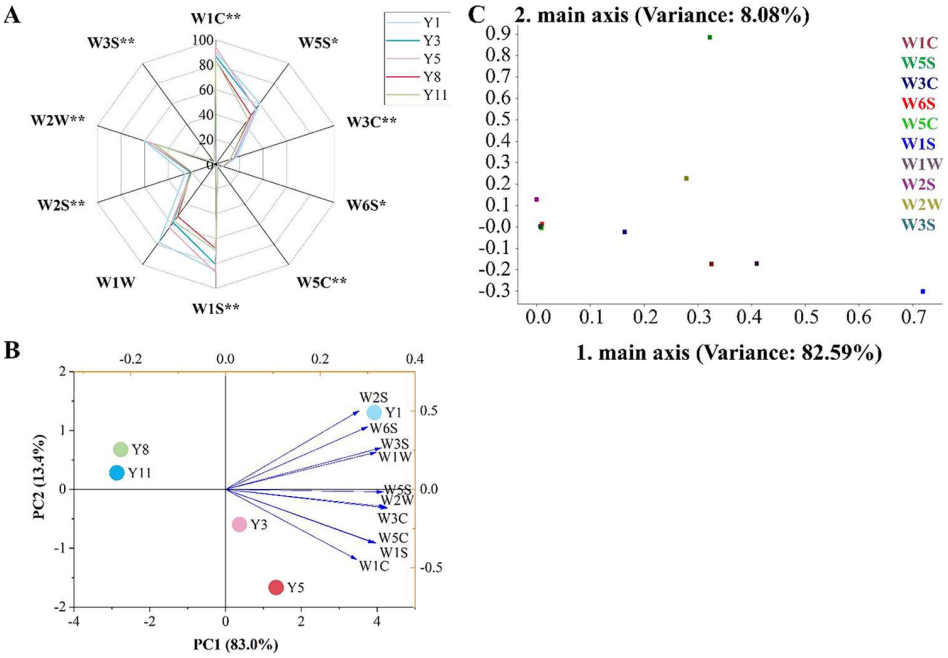


Figure 5. Results of E-nose analysis. (A) Radar map of electronic nose analysis. Statistical significance is given by (*) $p < 0.05$, and (**) $p < 0.01$. (B) principal component analysis. (C) Loadings analysis.

OAV, showing its important function, showing that the aroma intensity of a volatile compound was not exactly equal to the importance of taste. Furfural (Baked) was a special aroma compound in Y1. 1-nonanol (floral, fat) and isocaproic acid (fat) were aroma compounds specific to Y3, 2-nonanol (floral, fruity) was aroma compounds specific to Y5, 1-octene-3-ol (mushroom), dimethyl trisulfide (cabbage), 2-octanone (dairy, cheese), guaiacol (scented oil), 4-hydroxy-3-methoxystyrene (smoky), heptanal (aldehydic), undecan-2-one (fruit, wax), linalool (citrus, orange, lemon, floral), 3-octanone (fumes, aldehyde), and eugenol (clove, woody) were the main contributors to the Y11 characteristic aroma, although dimethyl trisulfide (cabbage) had a low threshold, it had a high odor activity value (775.06), consequently, it significantly impacts the overall aroma characteristics of the baijiu.[49]

To improve screening accuracy, more rigorous criteria (OAV ≥ 1, VIP > 1 and $P \leq 0.05$) were adopted to forecast potential key aroma compounds. The results showed that ethyl hexanoate, acetal, ethyl acetate, ethyl butanoate, 1-hexanol, pentanol, ethyl lactate, and 2-butanol were identified as 8 potential key aroma compounds, among which ethyl lactate and 2-butanol may be the key compounds causing the flavor differences between Y1 and other samples, and pentanol might be the key compound responsible for the flavor disparities between Y11 and other samples. Understanding their dynamics can contribute to improving the aroma quality of NXX baijiu during storage.

Comparison of Detection Results by GC-IMS and GC × GC-MS. There existed certain disparities in the detection capabilities for volatile compounds between GC-IMS and GC × GC-MS. The number of volatile compounds identified by GC-IMS (83) was less than that identified by GC × GC-MS (119). The main volatile compounds detected by GC-IMS were esters (37.33−41.16%) and alcohols (22.78−28.05%). The flavor characteristics of Y11 showed significant differences when compared to those of the other samples. The main volatile compounds identified by GC × GC-MS were esters (66.56−87.25%) and aldehydes (2.47−18.28%). The flavor characteristics of Y3 showed significant differences when compared to those of other samples. Therefore, the integration of the two methods can be employed to explore the changes in volatile compounds during the storage of NXX baijiu.

GC-IMS identified 41 potential marker metabolites, 11 key aroma compounds, 12 aromatic compounds with ROAV ≥ 1, and based on the peak area of volatile compounds in each sample, 2 potential key aroma compounds were recognized, and found that 1-pentanal had the ability to clearly distinguish the flavor variances between Y1 and other samples. Based on the concentration of volatile compounds, 65 potential marker metabolites, 16 key aroma compounds, 59 aromatic compounds with an OAV of ≥ 1, and 8 potential key aroma compounds were detected through GC × GC-MS. Integrated with the findings of GC-IMS and GC × GC-MS, it was found that acetal, isobutyric acid, and ethyl butanoate were the jointly identified aromatic compounds, and ethyl butanoate was the potential key aroma compound.

E-Nose Analysis. The E-nose showed high sensitivity to the flavor signatures present in the samples. Moreover, the intensity of the sensor response of the E-nose showed a positive correlation with the concentration of the corresponding volatile compounds in each sample.[50] As depicted in Figure 5A, W1C, W1S, W1W, and W5S sensors exhibited a higher reaction intensity, indicating that the contents of aromatics, methane, sulfide, terpenoids, and nitrogen oxides in each sample were higher. In Y5, the response intensities of W1C, W3C, W5C, and W1S sensors were greater than those in other years. The response intensity of W5S, W6S, W1W, W2S, W2W, and W3S sensors were the highest in Y1. In addition, the intensity of the W5S and W2W sensor response gradually decreased with increasing years.[51] The PCA score chart of response intensity (Figure 5B) showed that the two principal components were able to explain 96.4% of the total variance,

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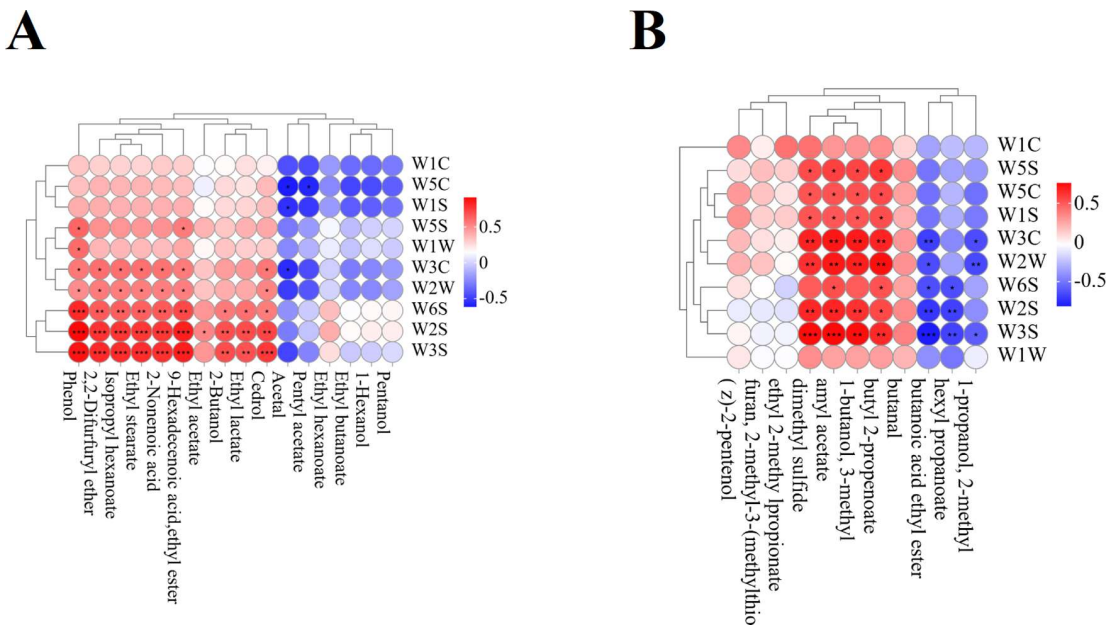


Figure 6. The heat map of the correlation between the E-nose sensor's response value and the levels of key aroma compound. (A) The heat map of the correlation between the E-nose sensor's response value and the levels of 16 key aroma compounds identified by GC×GC-MS. (B) The heat map of the correlation between the E-nose sensor's response value and the levels of 11 key aroma compounds identified by GC-IMS. The symbols "*" and "**" signify significant ($p < 0.05$) and extremely significant ($p < 0.01$) correlations, respectively. Positive ($0 < r < 0.5$) and negative ($-0.5 < r < 0$) correlations are displayed in red and blue, respectively.

indicating that the E-nose had a good ability to distinguish between different samples. Y3 and Y5 were close together; Y8 and Y11 were close together; and Y1 was located at a substantial distance away from the remaining samples. This indicated that the aroma distribution between Y3 and Y5 was more similar, that between Y8 and Y11 was more similar, and that the roasted, nutty, fruity, and ester flavors were more prominent in Y1, making it a unique style difference from other samples. Figure 5C shows the Loadings analysis of sensor response values for baijiu samples of different storage years. A total of 90.67% of the variance could be accounted for by the two principal components. The contribution of sensors W5S and W2W to 1 main axis was greater than that of 2 main axes, and the contribution of W2S to 2 main axEs was greater because the greater the load factor of the sensor, the larger the contribution to the overall response of the array.[52] W6S, W5C, and W3S had similar load factors and could be represented by a sensor.[53]

As presented in Figure 6, correlations were analyzed between the response values of the E-nose sensor and the levels of key aroma compounds, which were detected by GC-IMS and GC × GC-MS, respectively. As shown in Figure 6A, W3S, W2S, W6S, W2W, and W3C were associated with phenol, 2,2-difurfuryl ether, isopropyl hexanoate, ethyl stearate, 2-nonenoic acid, 9-hexadecenoic acid, ethyl ester, and cedrol were positively correlated, and W3S, W2S, and W6S were positively correlated with 2-butanol and ethyl lactate. There was a positive correlation between ethyl acetate and W2S, as well as a positive correlation between W1W and phenol, W5S was positively correlated with phenol and 9-hexadecenoic acid and ethyl ester. W3C, W1S, and W5C were negatively correlated with acetal, and W5C was negatively correlated with pentyl acetate. According to the GC × GC-MS result, 2,2-difurfuryl ether, isopropyl hexanoate, and ethyl stearate were only found in Y1. This indicated that the E-nose can effectively distinguish Y1 from other baijiu samples based on its specific response to these key aroma compounds. These findings were in line with the PCA outcomes of the E-nose analysis. As presented in Figure 6B, W3S, W2S, W2W, W3C, W1S, W5C, and W5S and dimethyl sulfide, amyl acetate, 1-butanol, 3-methyl, and butyl 2-propenoate were positively correlated, W6S was positively correlated with amyl acetate and butyl 2-propenoate, W3S, W2S, W6S, W2W, and W3C were negatively correlated with butanoic acid ethyl ester, W3S,W2S, and W6S were negatively correlated with hexyl propanoate, W3S, W2W, and W3C were negatively correlated with 1-propanol, 2-methyl.

In summary, in this study, GC-IMS, GC × GC-MS, and E-nose were utilized to conduct a comprehensive analysis of the volatile compounds in NXX baijiu with 5 different storage years (Y1, Y3, Y5, Y8, and Y11). A total of 83 volatile compounds were detected by means of GC-IMS. Among them, the major volatile compounds were esters and alcohols. The concentration of esters was the highest in Y5, while the concentration of ketones was the highest in Y8, Y11 had obvious differences in flavor characteristics from other samples, and 1-pentanal could significantly distinguish the flavor differences between Y1 and the other samples. A total of 119 volatile compounds were detected by means of GC-MS. The variety of volatile compounds increased as the storage time was extended. Moreover, the flavor characteristics of Y3 were different from those of other samples. Esters, alcohols, acids, and aldehydes were most abundant in Y8, enhancing the overall aroma. Y11 had the highest diversity of esters, alcohols, phenolic ketones, and other volatile compounds. Additionally, it had the largest number of unique volatile compounds. Ethyl lactate and 2-butanol might be the crucial compounds responsible for the flavor disparity between Y1 and the other samples. Meanwhile, pentanol could potentially be the crucial compound responsible for the flavor disparity between Y11

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and the other samples. Combined with the findings of GC-IMS and GC × GC-MS, it was found that acetal, isobutyric acid, and ethyl butanoate were the jointly identified aromatic compounds, and ethyl butanoate was the potential key aroma compound. The results of E-nose analysis showed that Y1 was more prominent in roasting, nut, fruit, and ester aromas, which made it different from other samples. A strong correlation with particular key aroma compounds was demonstrated by some E-nose sensors, indicating that the E-nose was capable of effectively differentiating the NXX baijiu with different storage times according to its specific aroma characteristics. The findings of this research contribute to enhancing the understanding of flavor changes in NXX baijiu during different storage times, and provide important insights for flavor and quality evaluation of NXX baijiu.

However, the relationship between the specific aroma characteristic and volatile components remains to be further verified. To gain a more comprehensive understanding and optimize storage conditions to enhance the aroma quality of NXX baijiu, an in-depth and all-round investigation into storage temperature, storage methods, and analytical techniques will be required in the future. Additionally, revealing the mechanism of the complex evolution process of aroma compounds and their influence on the overall aroma of NXX baijiu will become an innovative research hotspot. These studies can not only scientifically explain the fundamental reasons for inducing volatile compounds and aroma changes but also be beneficial for controlling storage time and optimizing storage methods, thereby promoting the improvement of the aroma, taste, and quality of NXX baijiu during storage.

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Notes

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ABBREVIATIONS

NXX, Nongxiangxing; Y1, Nongxiangxing baijiu of 1 year; Y3, Nongxiangxing baijiu of 3 year; Y5, Nongxiangxing baijiu of 5 year; Y8, Nongxiangxing baijiu of 8 year; Y11, Nongxiangxing baijiu of 11 year; GC-IMS, gas chromatography-ion migration spectrometry; GC × GC-MS, two-dimensional gas chromatography-combined mass spectrometry; VIP, variable projected importance; ROAV, relative odor activity values; OAV, odor activity values; E-nose, electronic nose; OPLS-DA, orthogonal partial least-squares discriminant analysis; PCA, principal component analysis; RI, retention index; Rt, retention time; Dt, drift time

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